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(54) **SEMICONDUCTOR DEVICE STRUCTURE
WITH INNER SPACER LAYER**

(71) Applicant: **TAIWAN SEMICONDUCTOR
MANUFACTURING COMPANY,
LTD.**, Hsinchu (TW)

(72) Inventors: **Han-Yu Lin**, Nantou County (TW);
Chansyun David Yang, Shinchu (TW);
Fang-Wei Lee, Hsinchu (TW);
Tze-Chung Lin, Hsinchu (TW); **Li-Te
Lin**, Hsinchu (TW); **Pinyen Lin**,
Rochester, NY (US)

(73) Assignee: **TAIWAN SEMICONDUCTOR
MANUFACTURING COMPANY,
LTD.**, Hsinchu (TW)

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Mar. 13, 2023, now Pat. No. 11,973,129, which is a
(Continued)

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H01L 21/02 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H10D 30/0243** (2025.01); **H01L 21/0214**
(2013.01); **H01L 21/02167** (2013.01);
(Continued)

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CPC H01L 29/165
See application file for complete search history.

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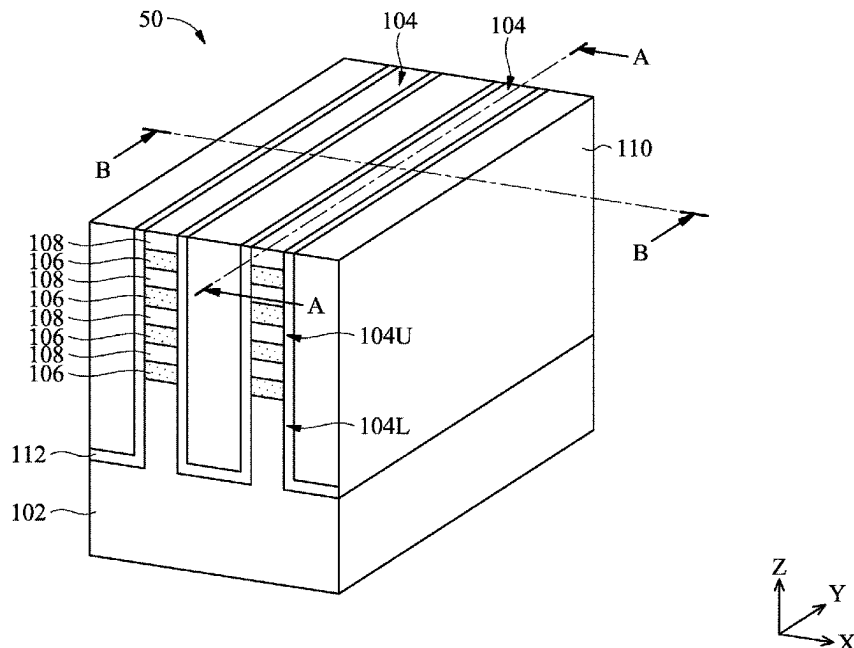
Primary Examiner — Mounir S Amer

(74) *Attorney, Agent, or Firm* — Birch, Stewart, Kolasch
& Birch, LLP

(57) **ABSTRACT**

A semiconductor device structure is provided. The semicon-
ductor device structure includes forming semiconductor
device structure includes a gate stack wrapping around a
plurality of nanowire structures. The gate stack includes a
first portion above the plurality of nanowire structures and
second portions between the nanowire structures. The semi-
conductor device structure further includes a gate spacer
layer along a sidewall of the first portion of the gate stack,
and a plurality of inner spacer layers along sidewalls of the
second portions of the gate stack. The gate spacer layer has
a first carbon concentration, the inner spacer layers have a
second carbon concentration, and the second carbon con-
centration is lower than the first carbon concentration.

20 Claims, 21 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/504,104, filed on Oct. 18, 2021, now Pat. No. 11,605,728, which is a division of application No. 16/299,531, filed on Mar. 12, 2019, now Pat. No. 11,152,491.

- (60) Provisional application No. 62/721,931, filed on Aug. 23, 2018.

(51) **Int. Cl.**

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H01L 21/321 (2006.01)
H01L 21/768 (2006.01)
H10D 30/43 (2025.01)
H10D 30/62 (2025.01)
H10D 30/67 (2025.01)
H10D 62/10 (2025.01)
H10D 62/822 (2025.01)
H10D 64/01 (2025.01)
H10D 84/01 (2025.01)
H10D 84/03 (2025.01)

(52) **U.S. Cl.**

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H10D 30/6735 (2025.01); **H10D 30/6757** (2025.01); **H10D 62/121** (2025.01); **H10D 62/822** (2025.01); **H10D 64/017** (2025.01); **H10D 64/018** (2025.01); **H10D 84/0128** (2025.01); **H10D 84/013** (2025.01); **H10D 84/0147** (2025.01); **H10D 84/038** (2025.01)

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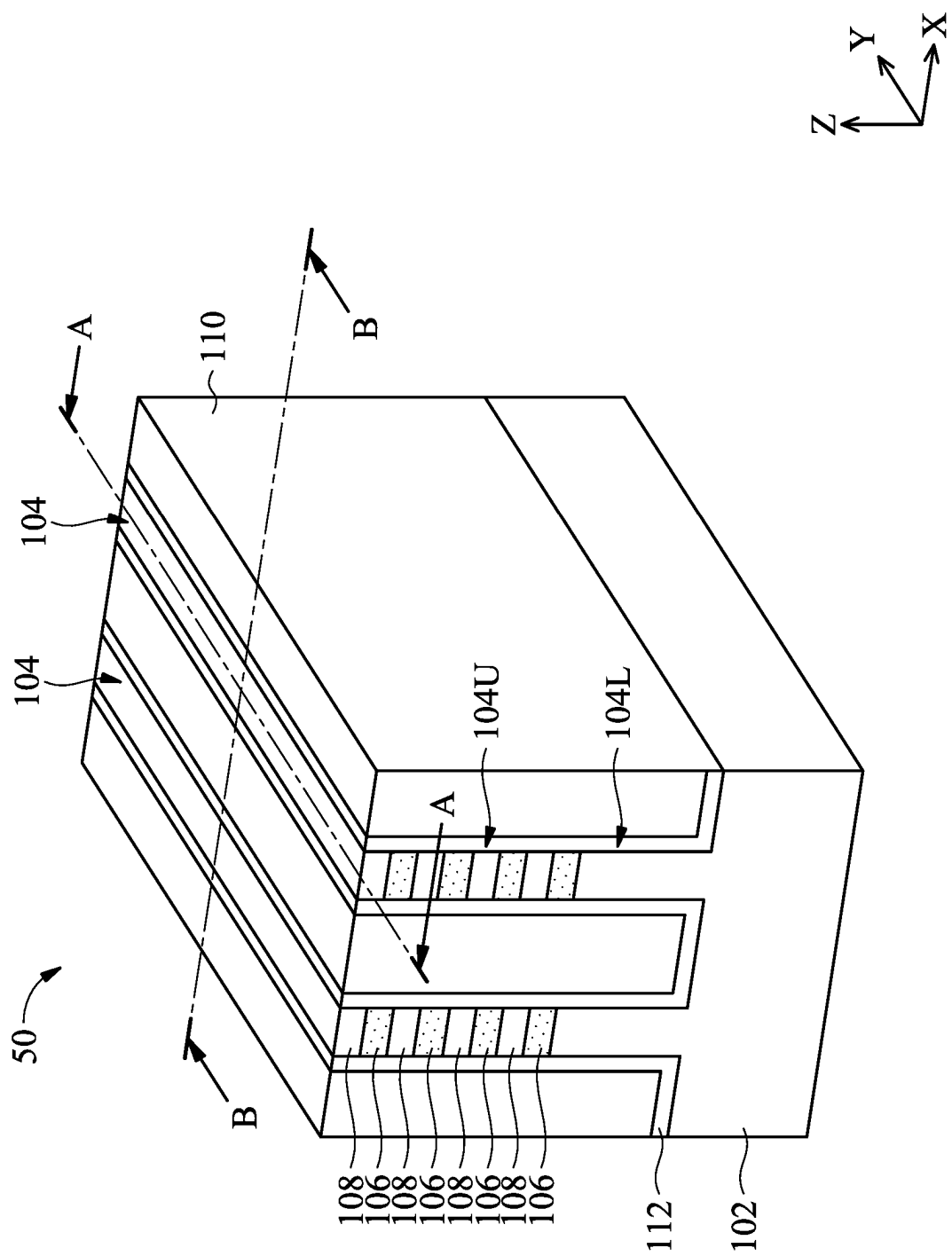


FIG. 1

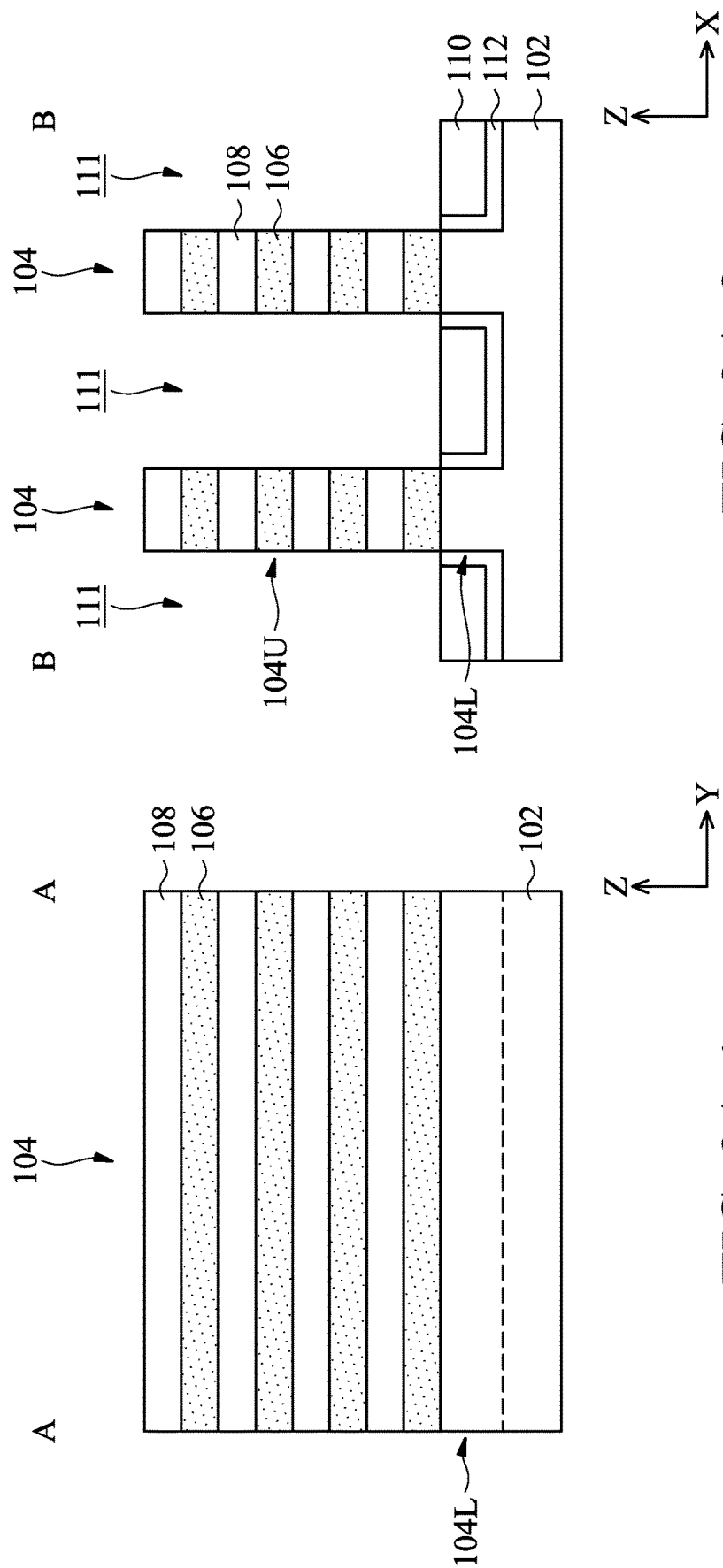


FIG. 2A-2

FIG. 2A-1

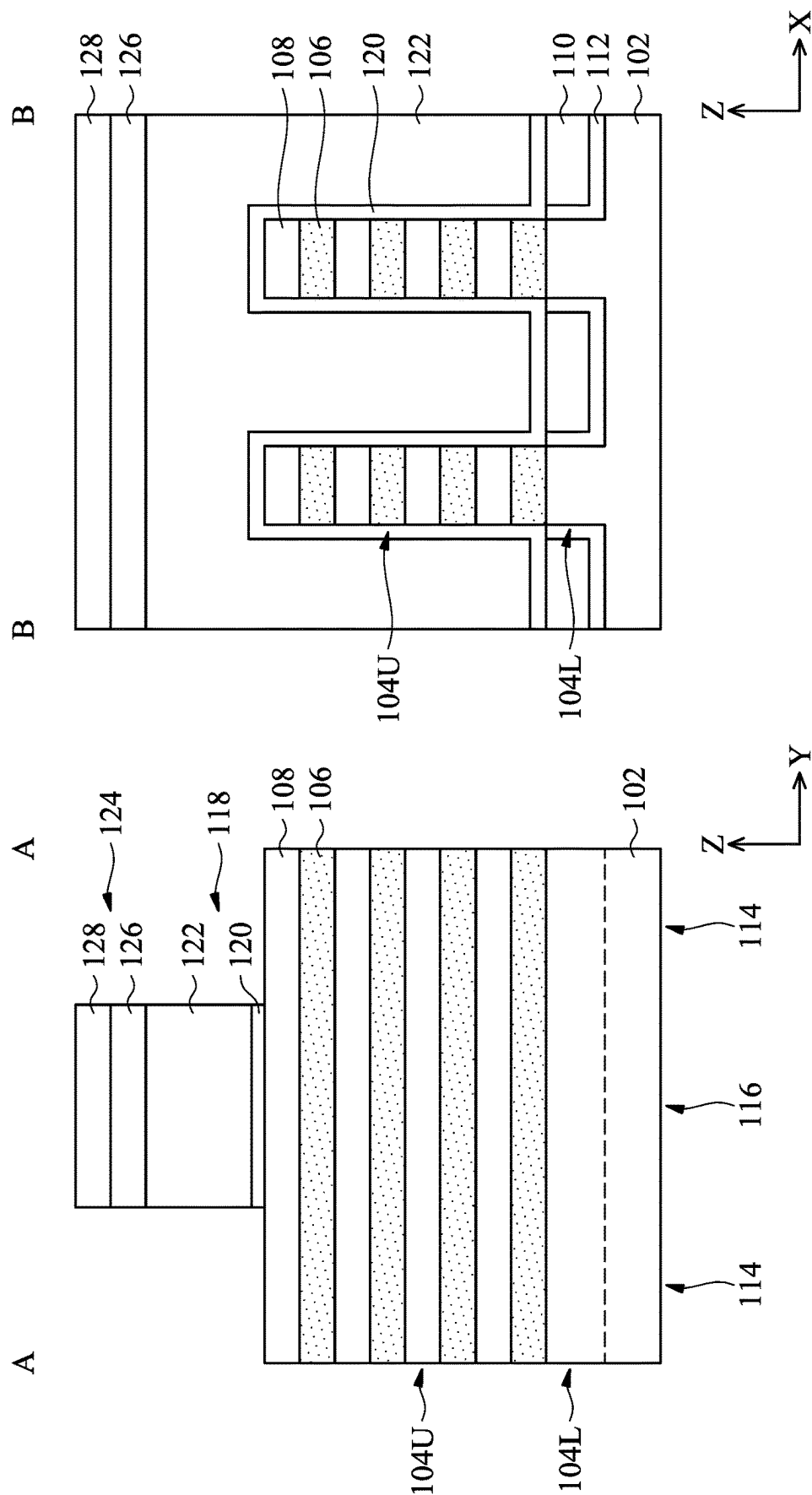
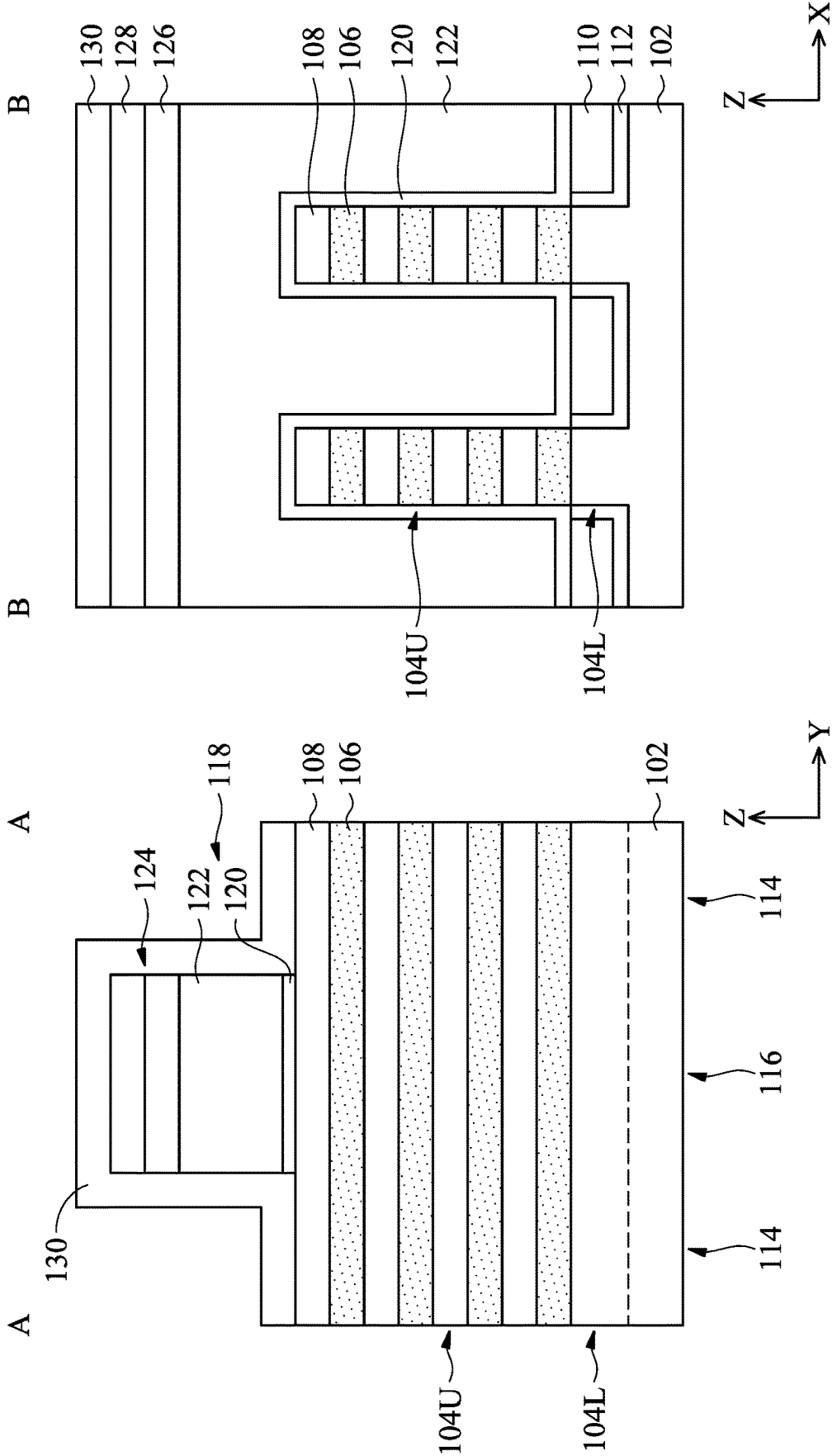


FIG. 2B-2

FIG. 2B-1



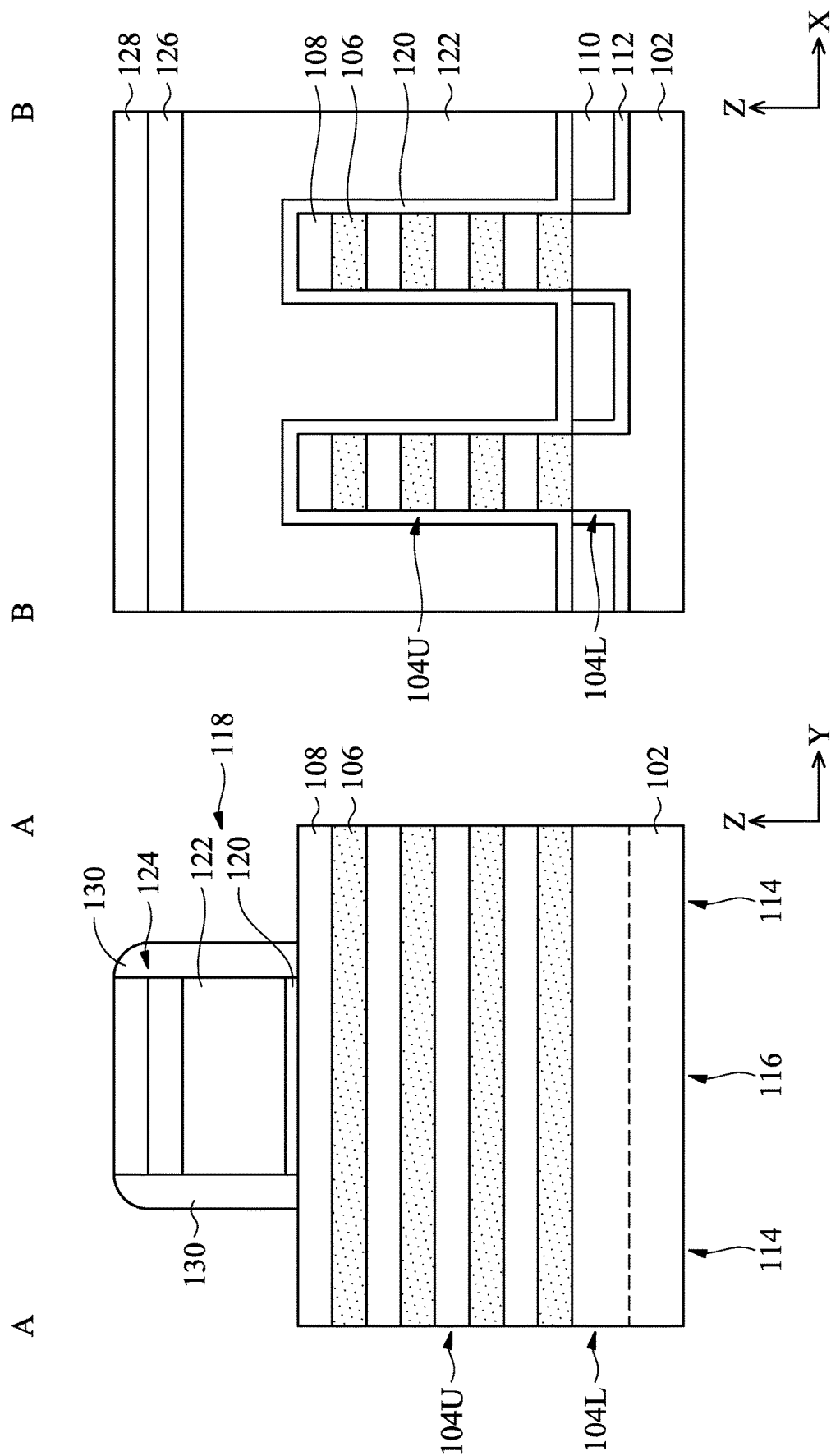


FIG. 2D-2

FIG. 2D-1

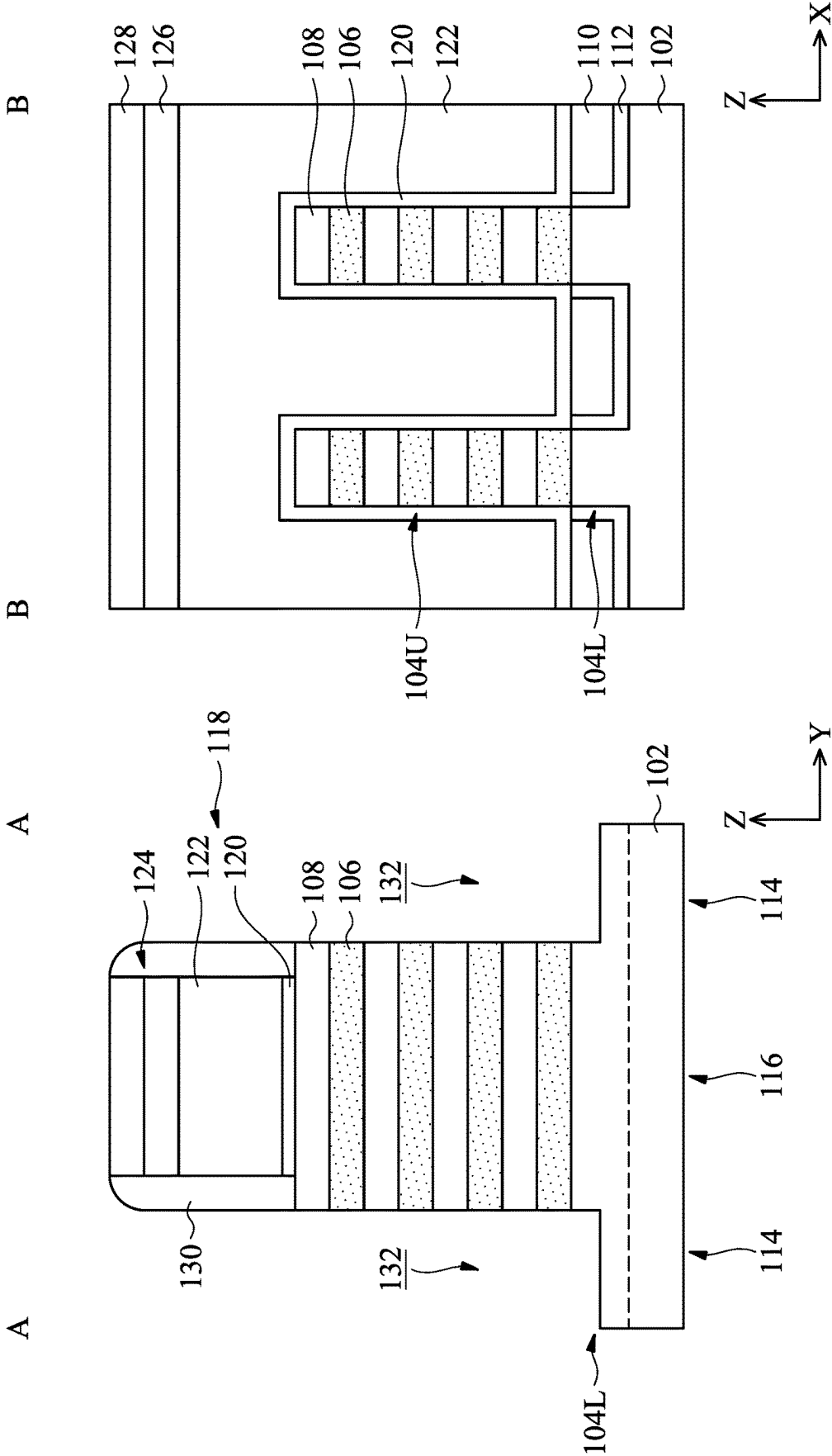


FIG. 2E-1

FIG. 2E-2

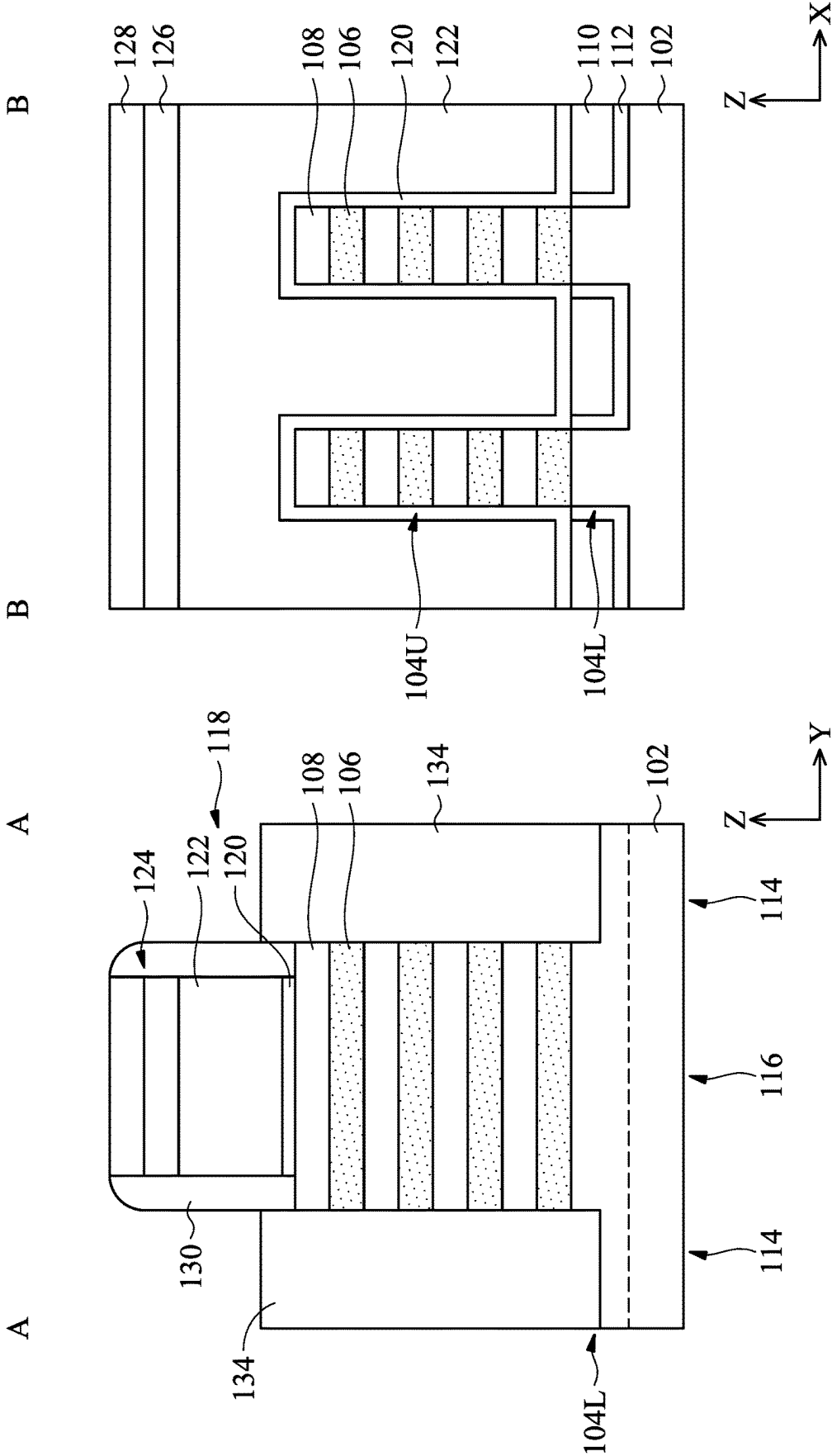


FIG. 2F-1

FIG. 2F-2

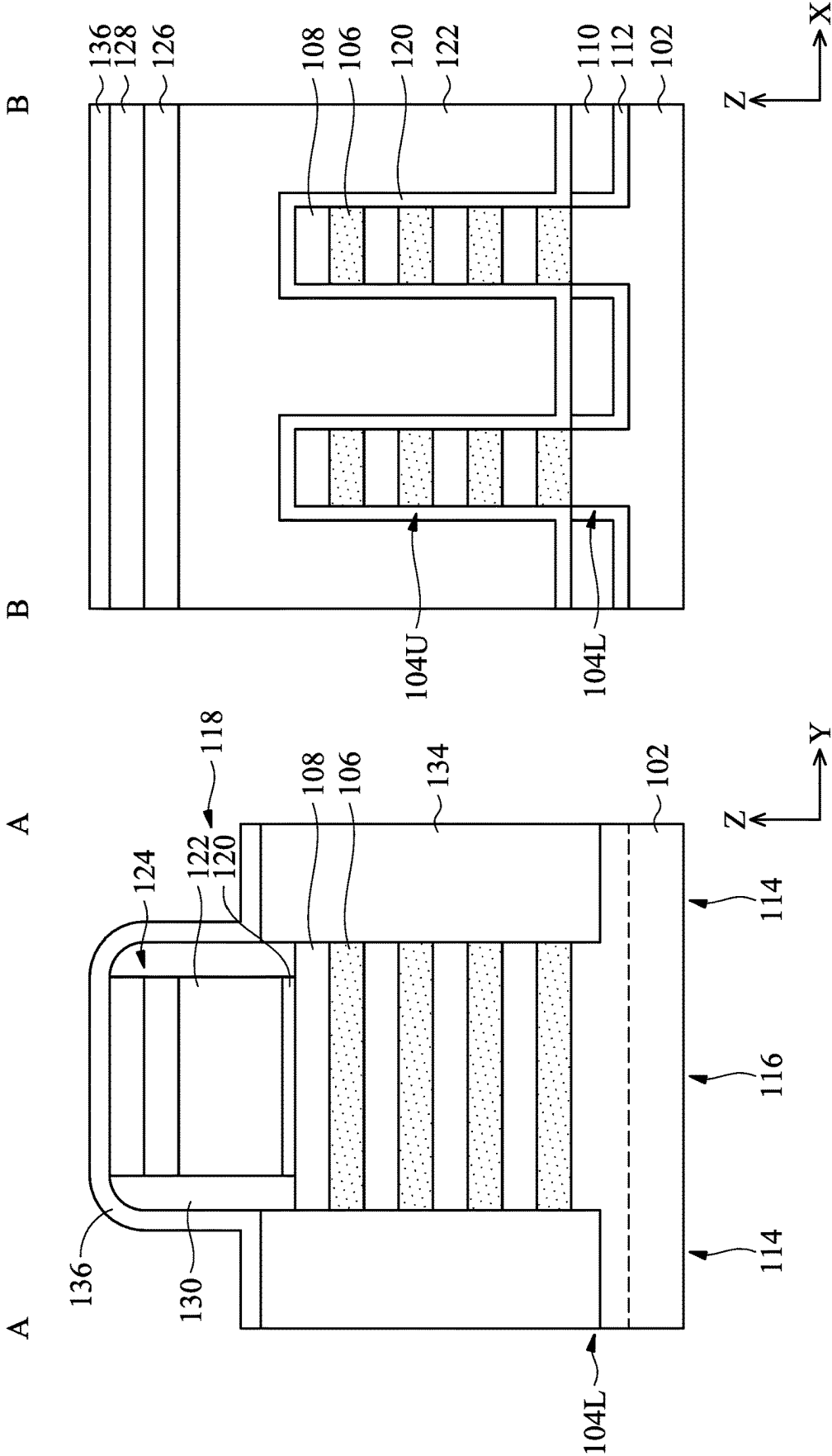
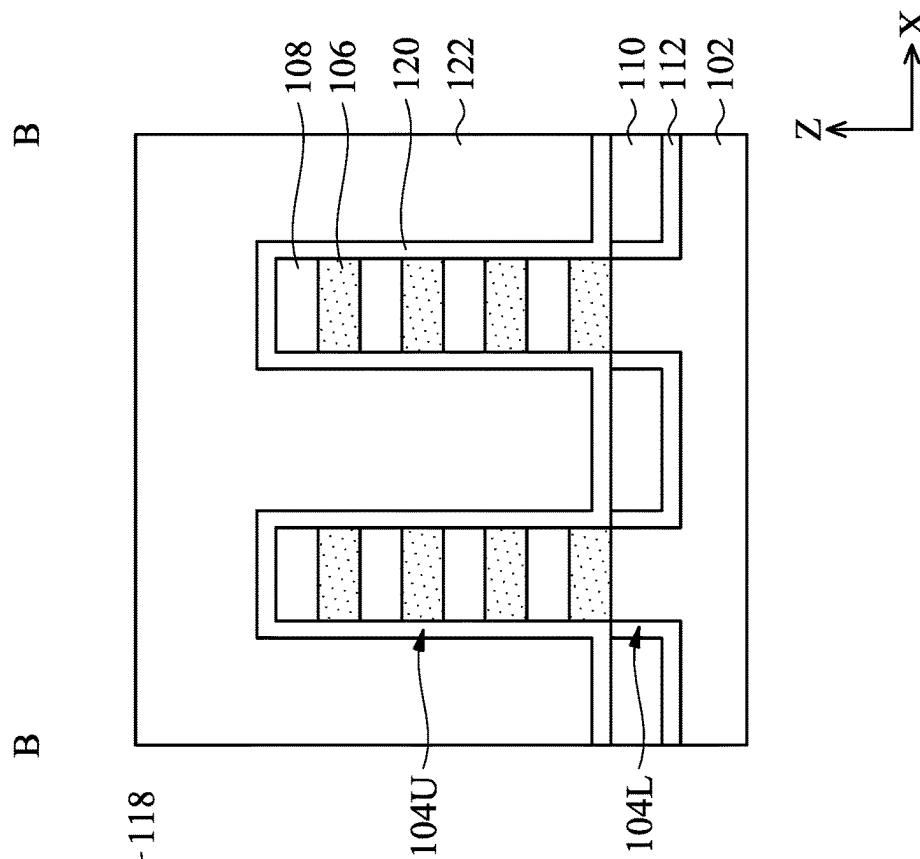
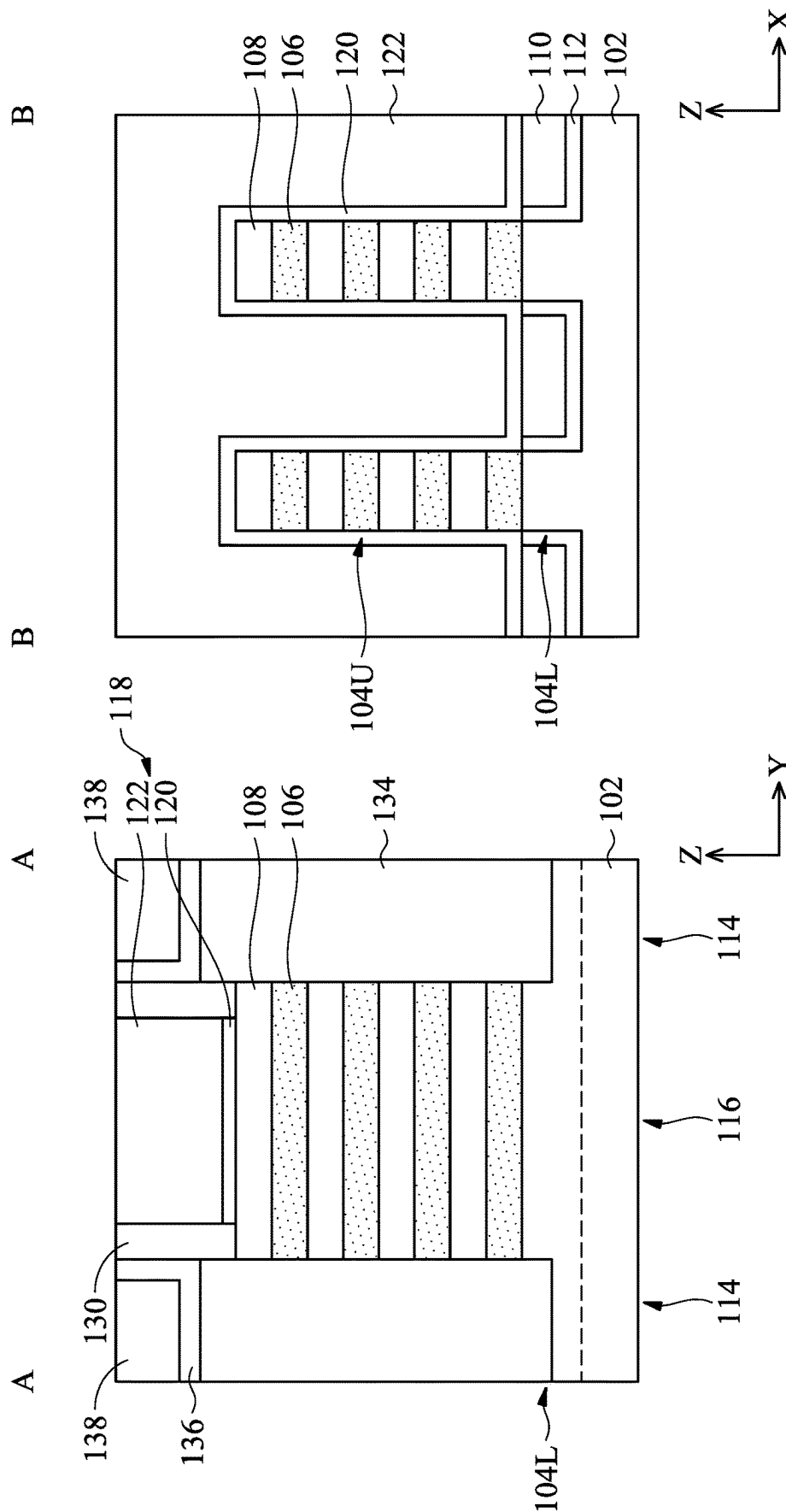
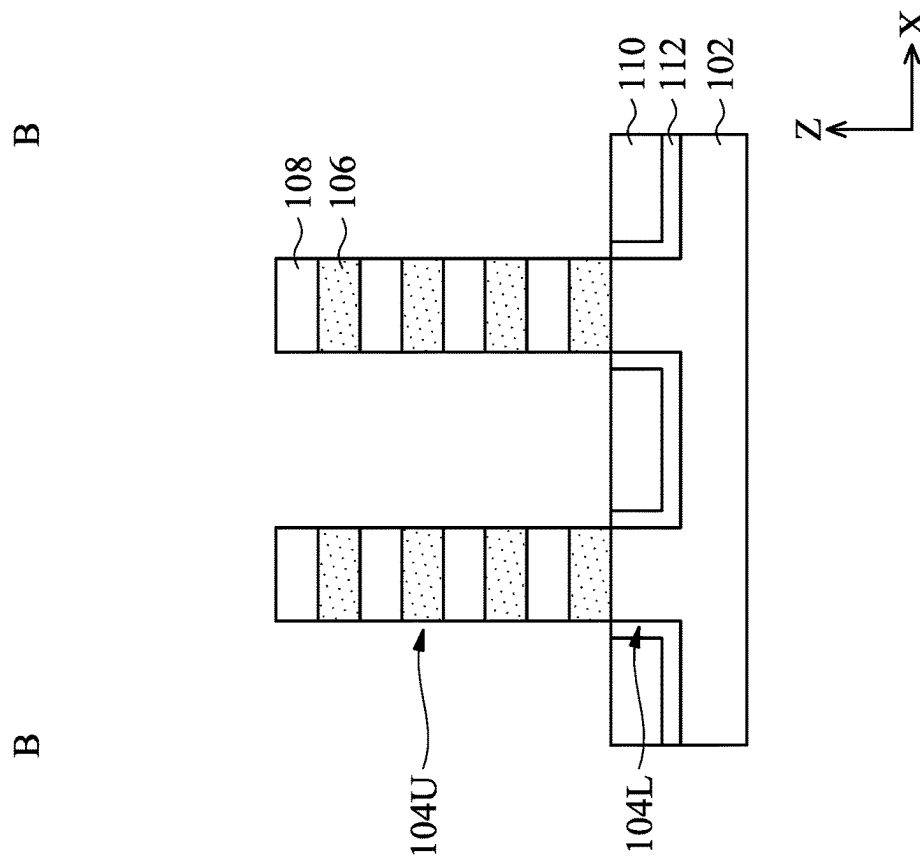
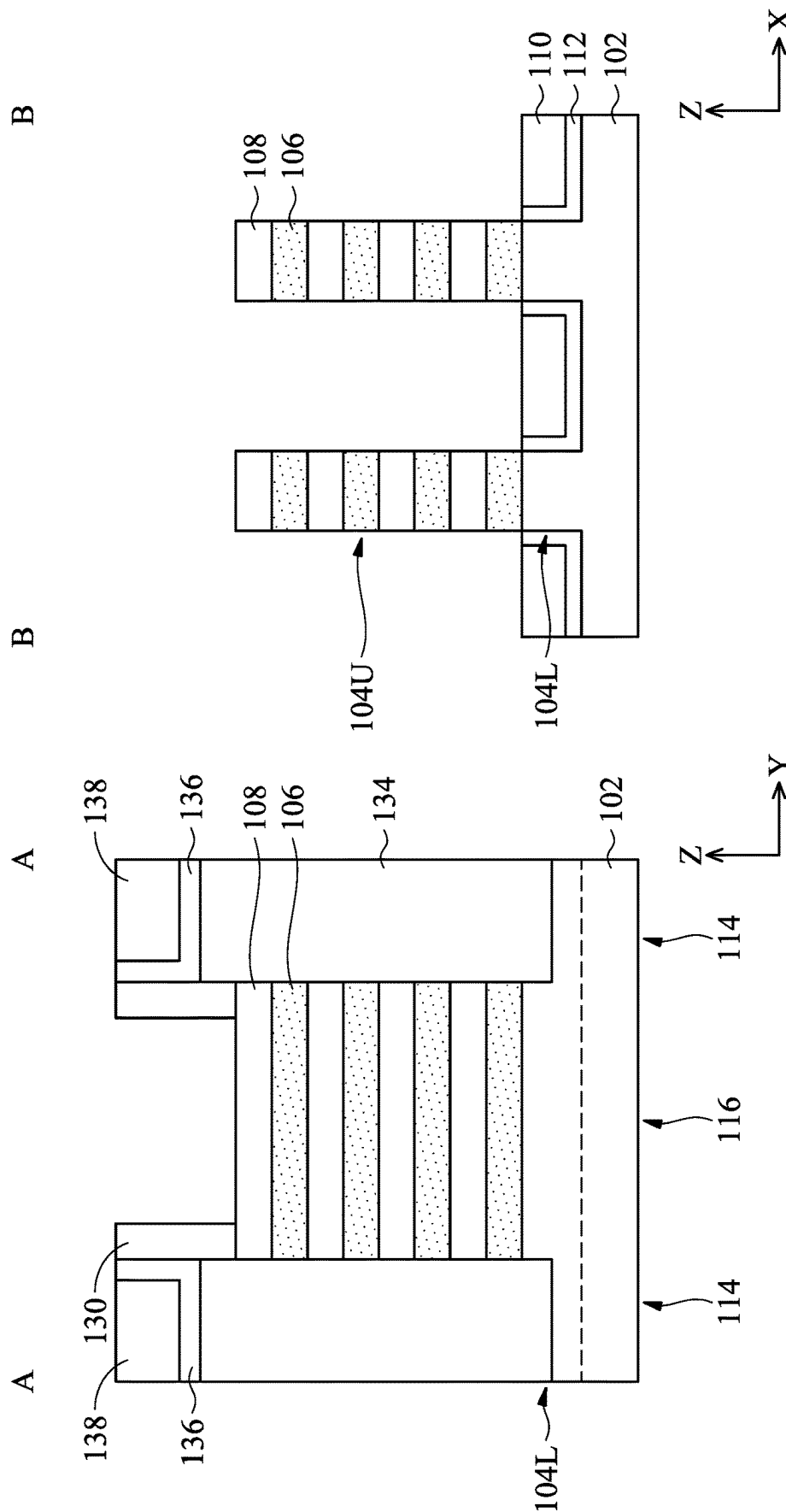


FIG. 2G-2

FIG. 2G-1





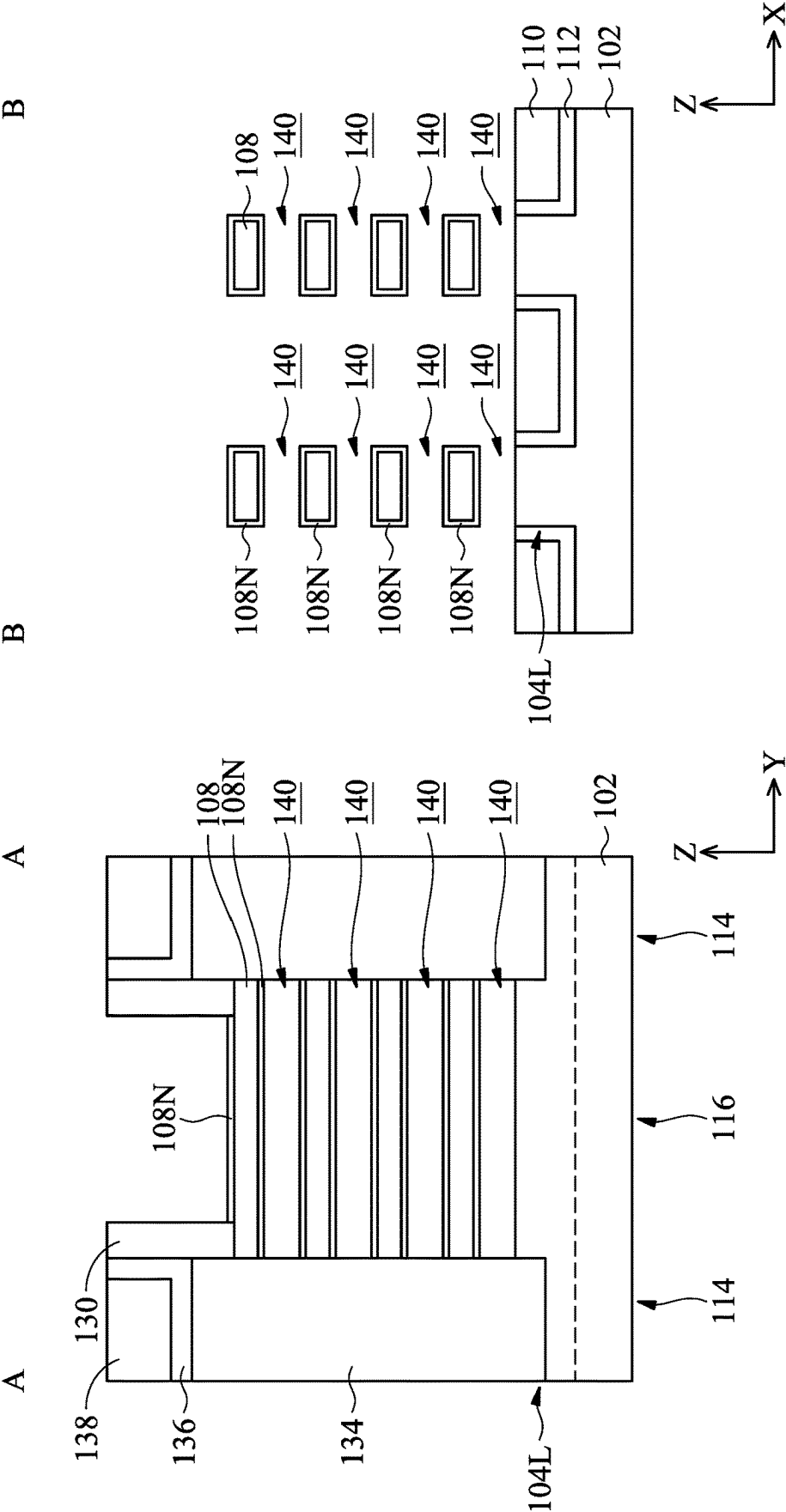


FIG. 2J-2

FIG. 2J-1

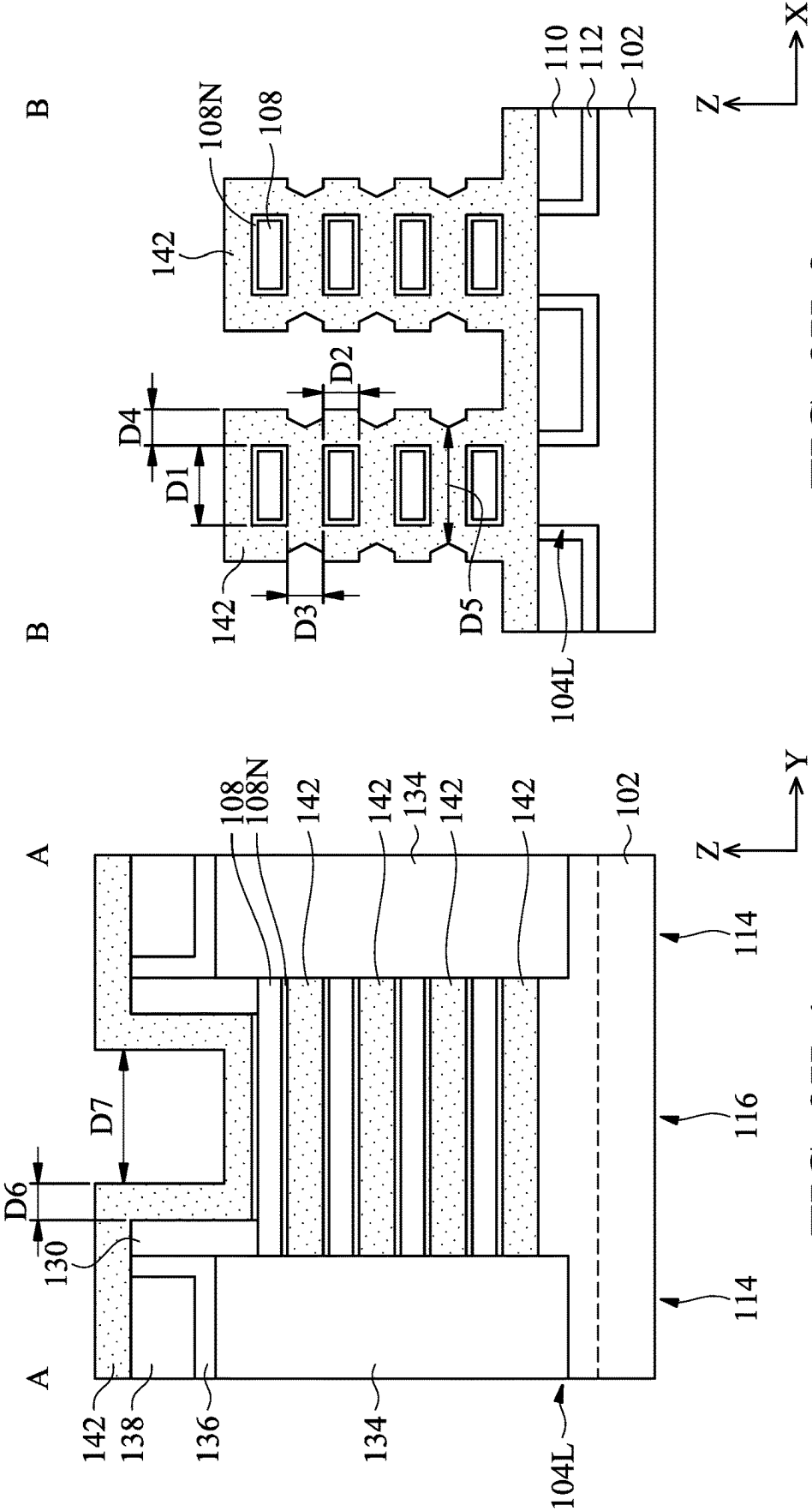


FIG. 2K-1

FIG. 2K-2

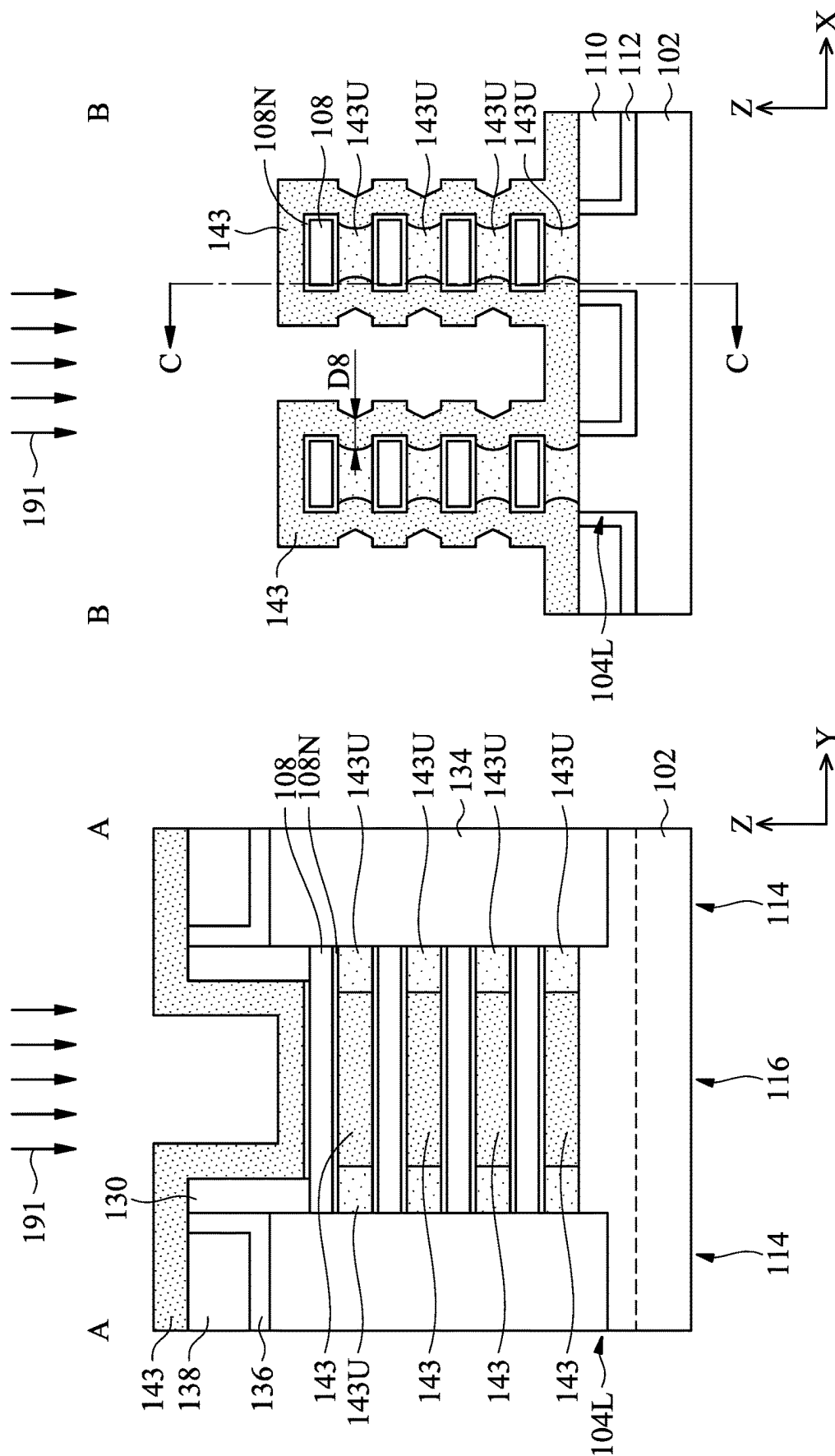


FIG. 2L-2

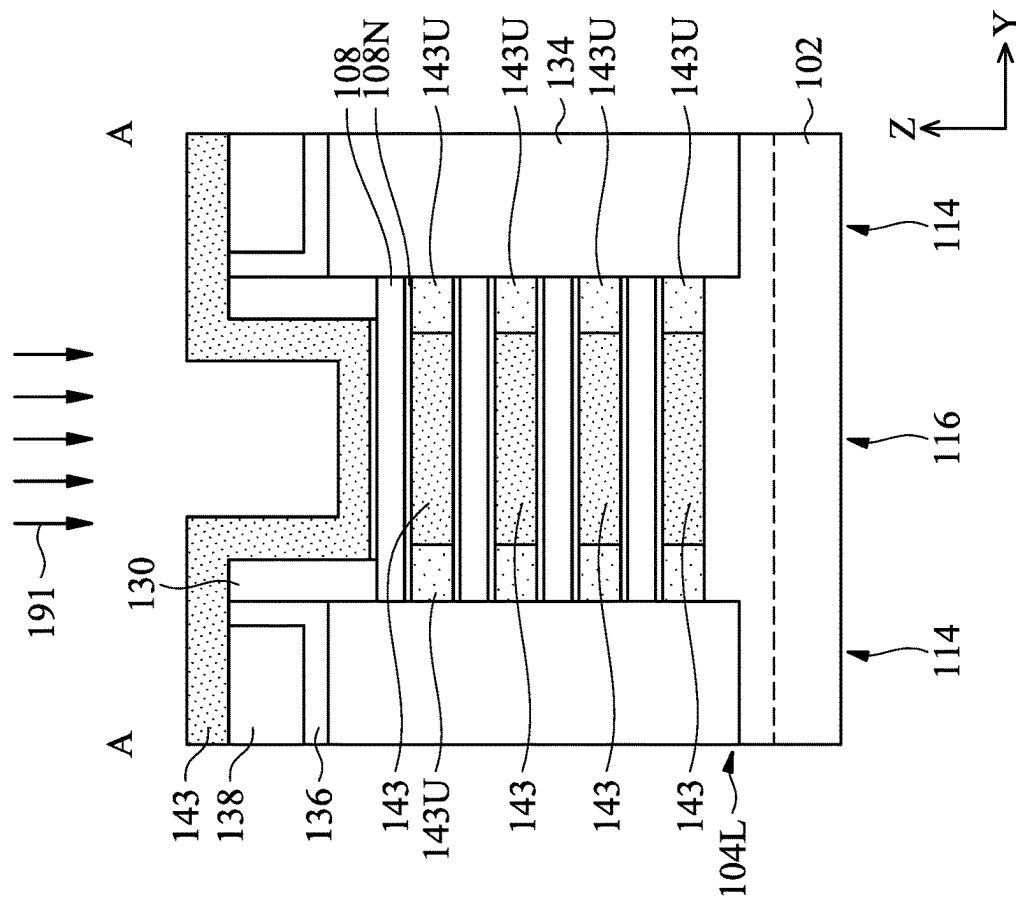


FIG. 2L-1

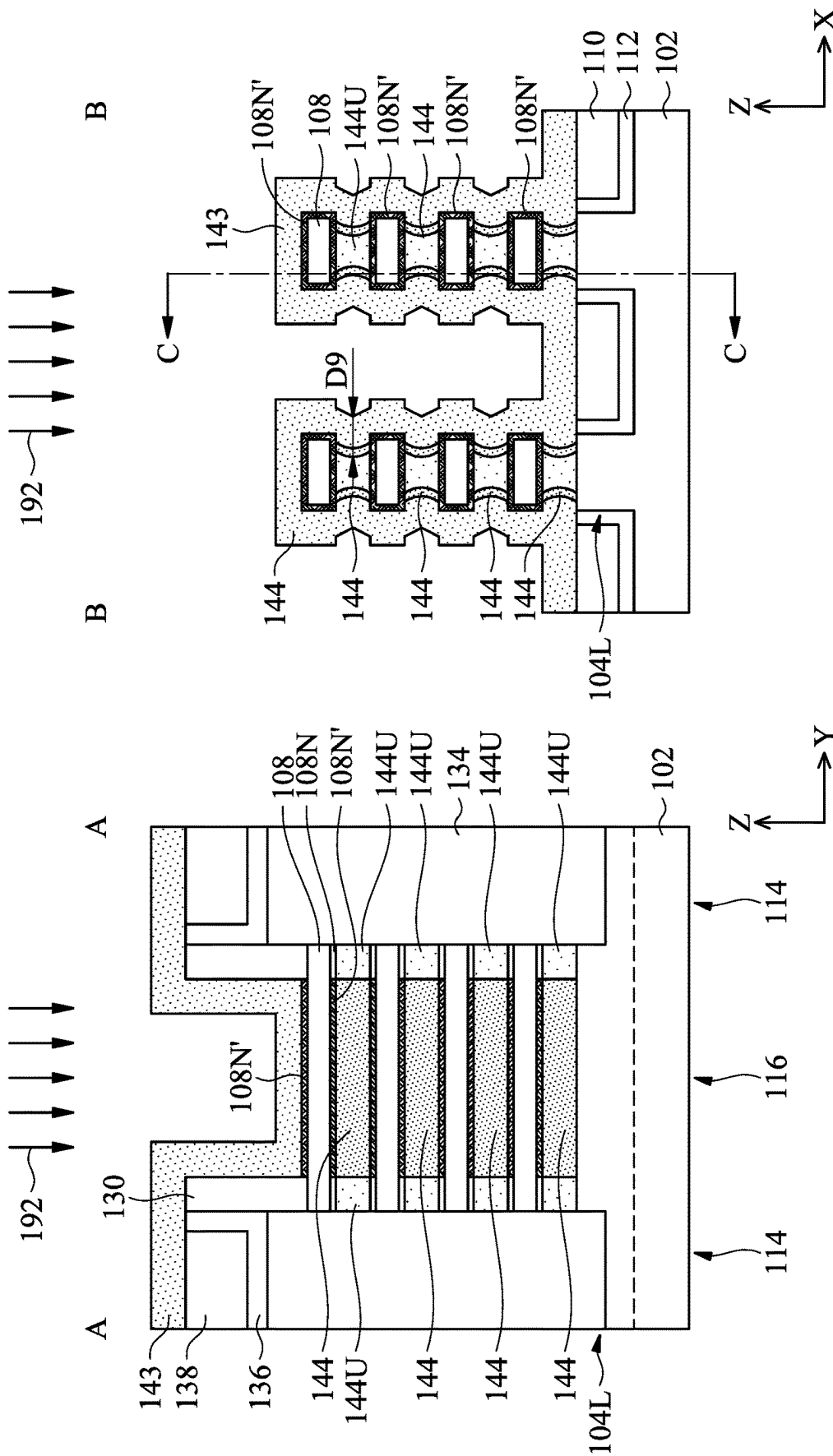


FIG. 2M-2

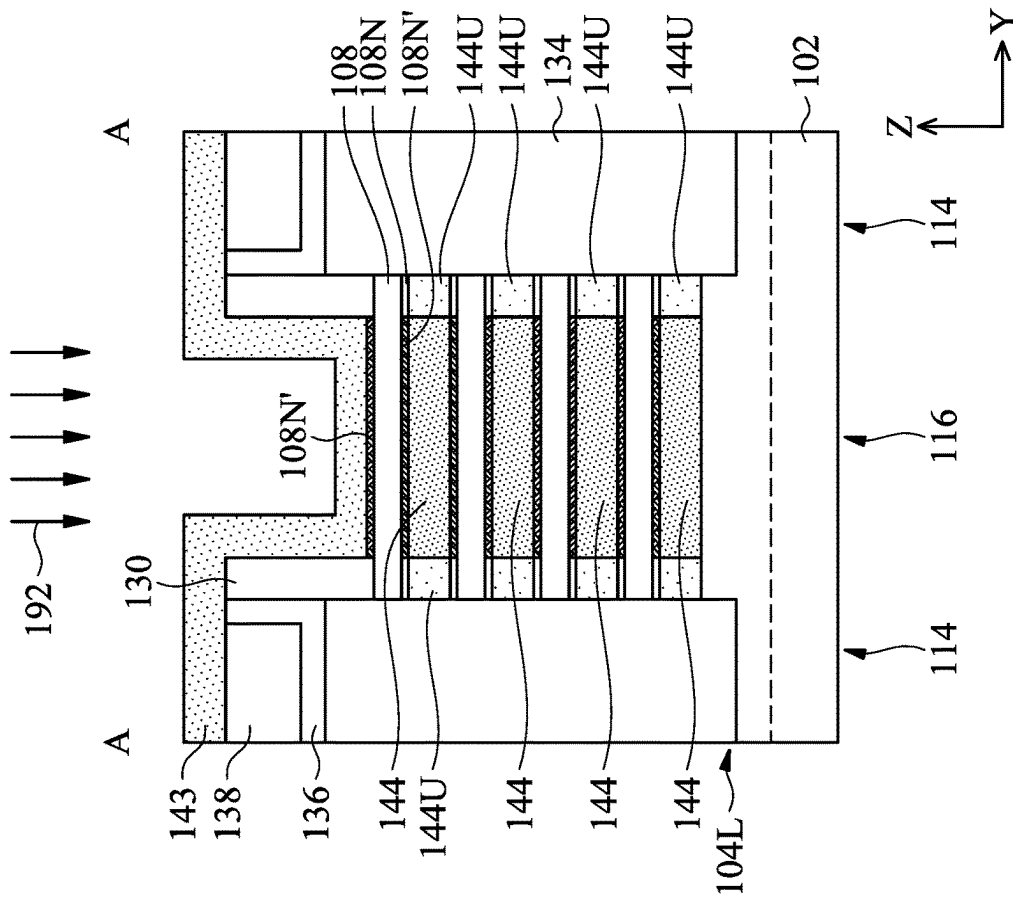


FIG. 2M-1

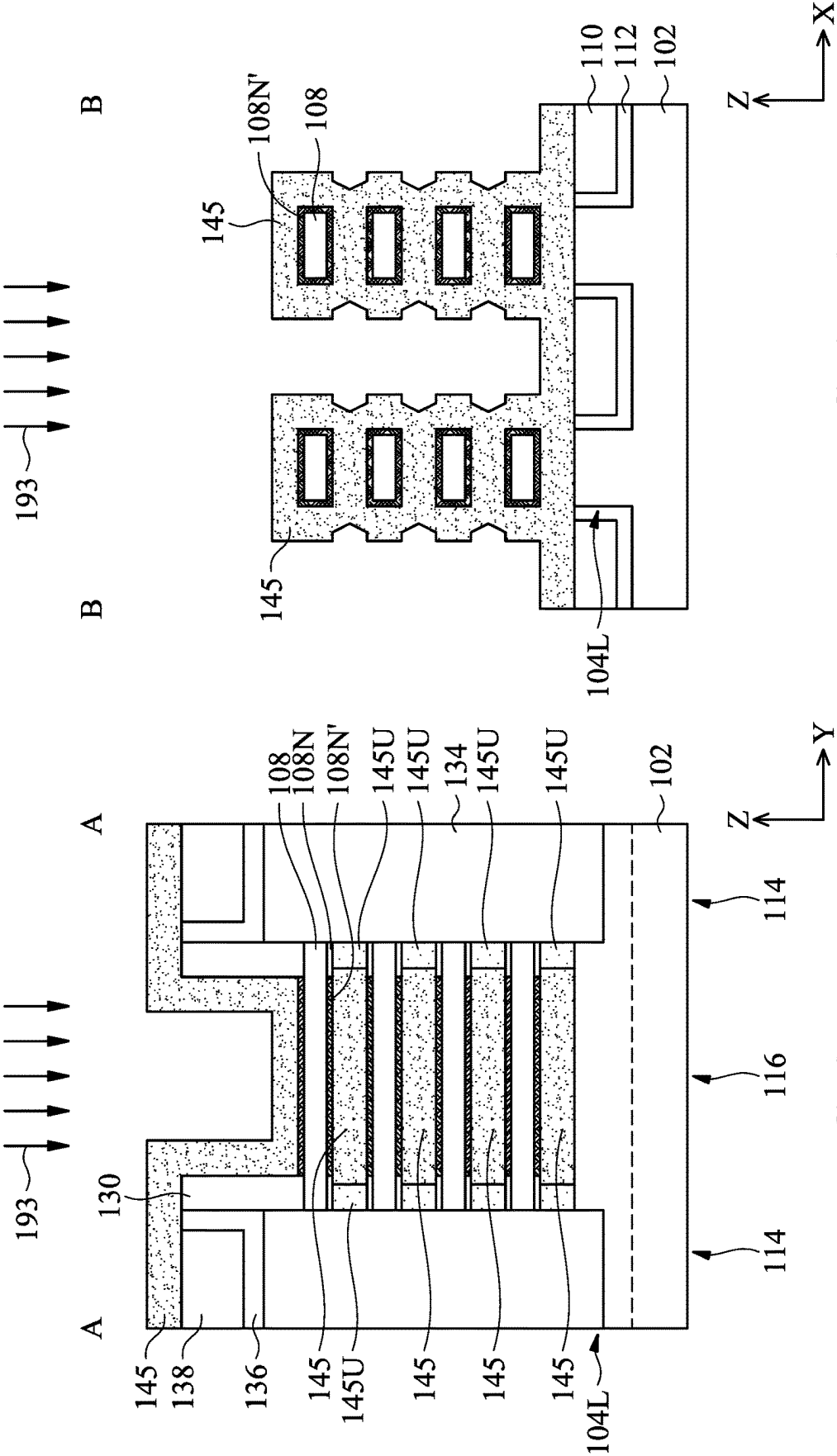


FIG. 2N-2

FIG. 2N-1

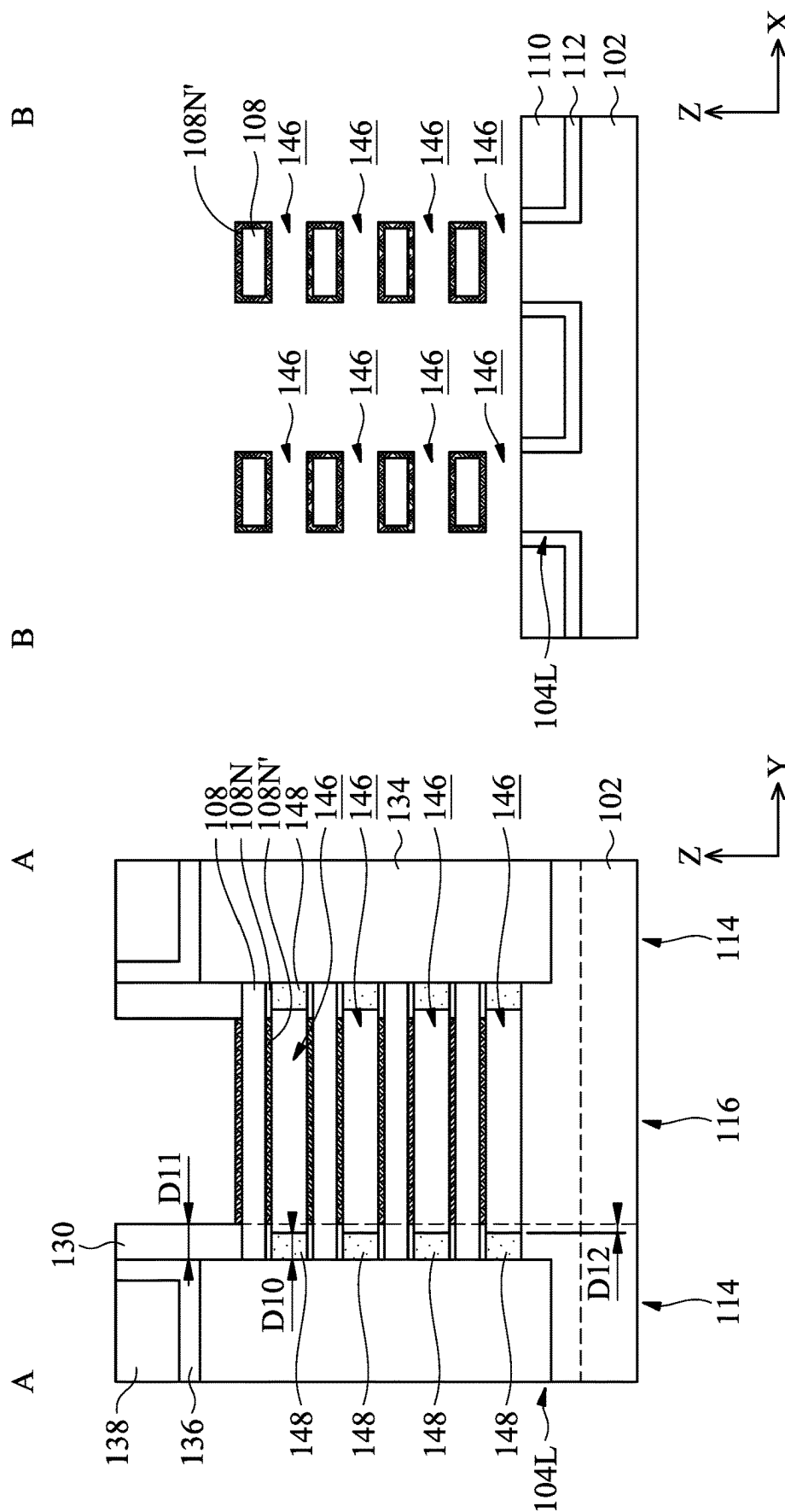


FIG. 20-2

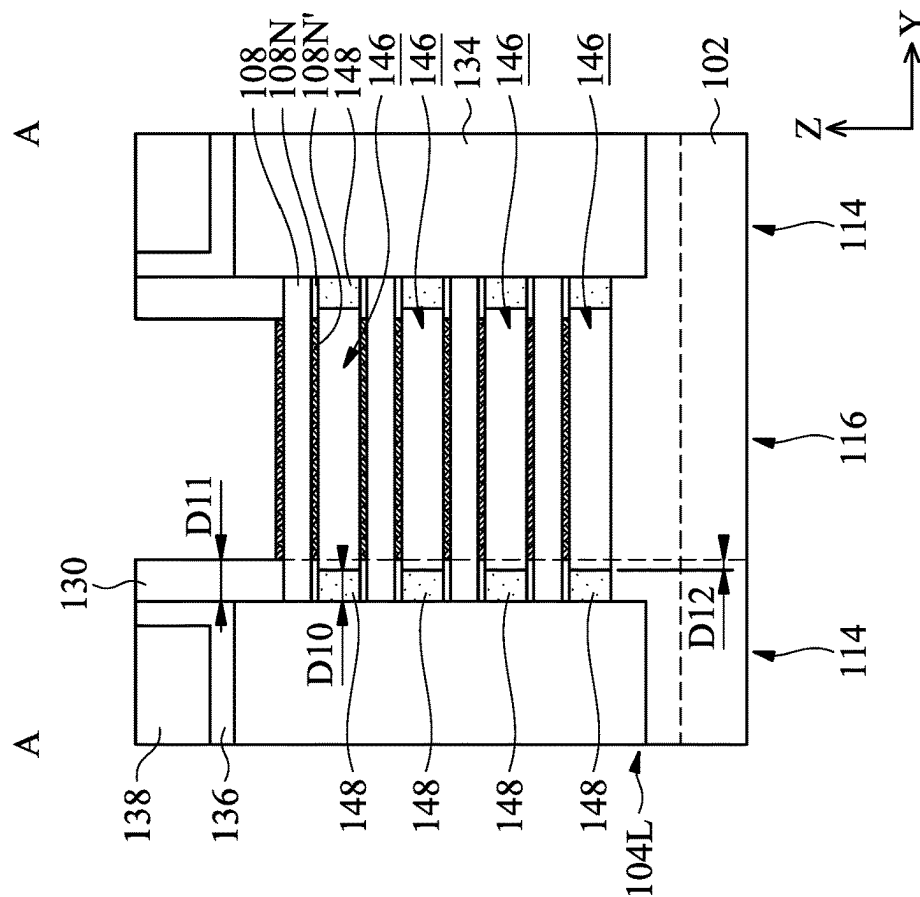


FIG. 20-1

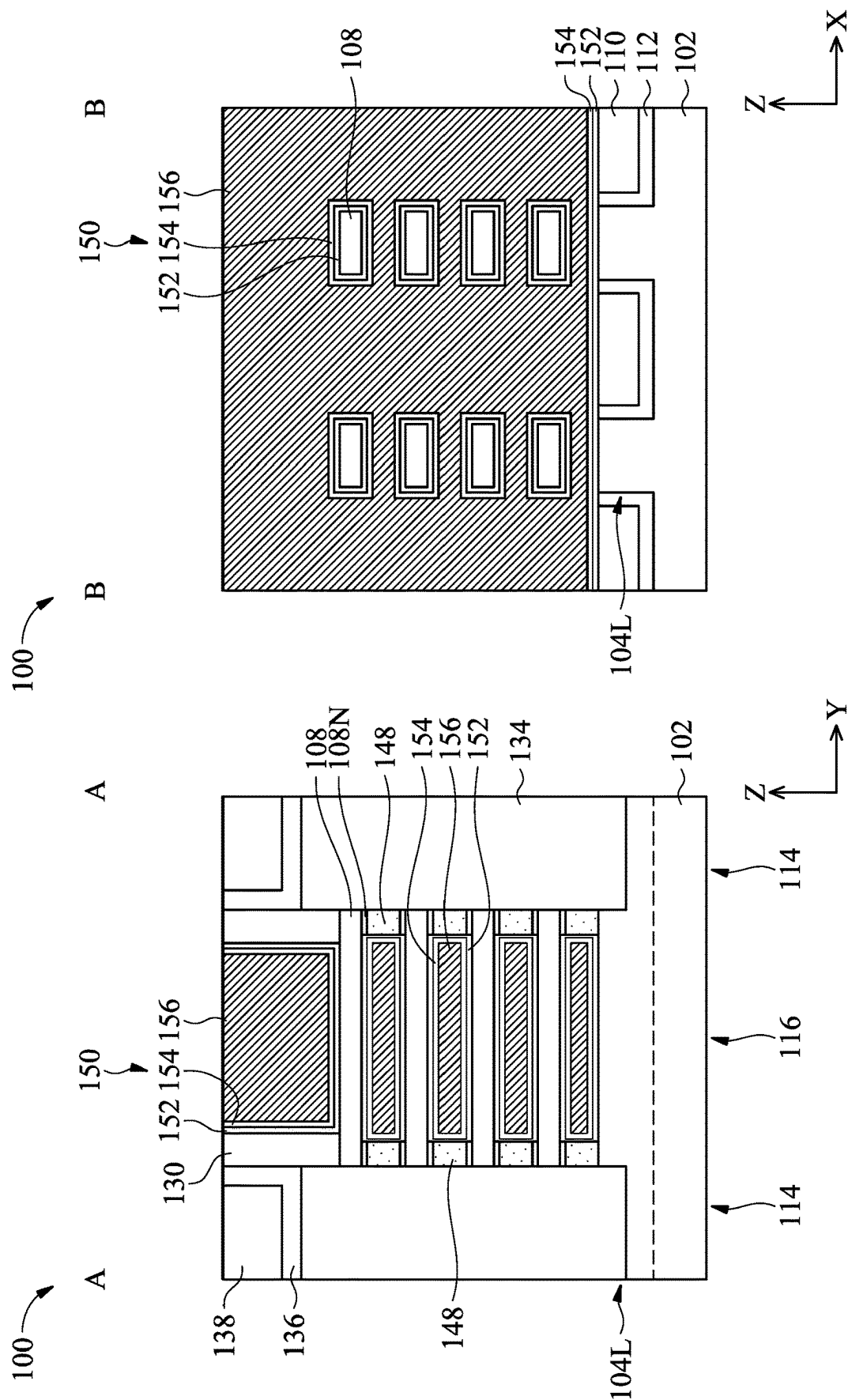
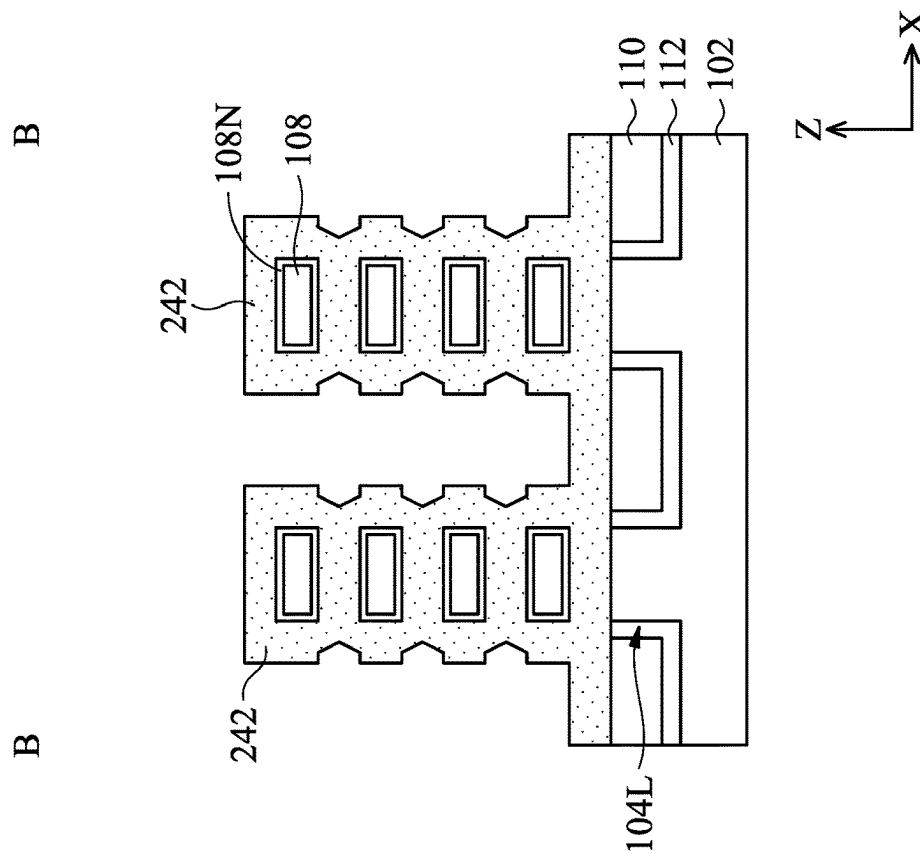
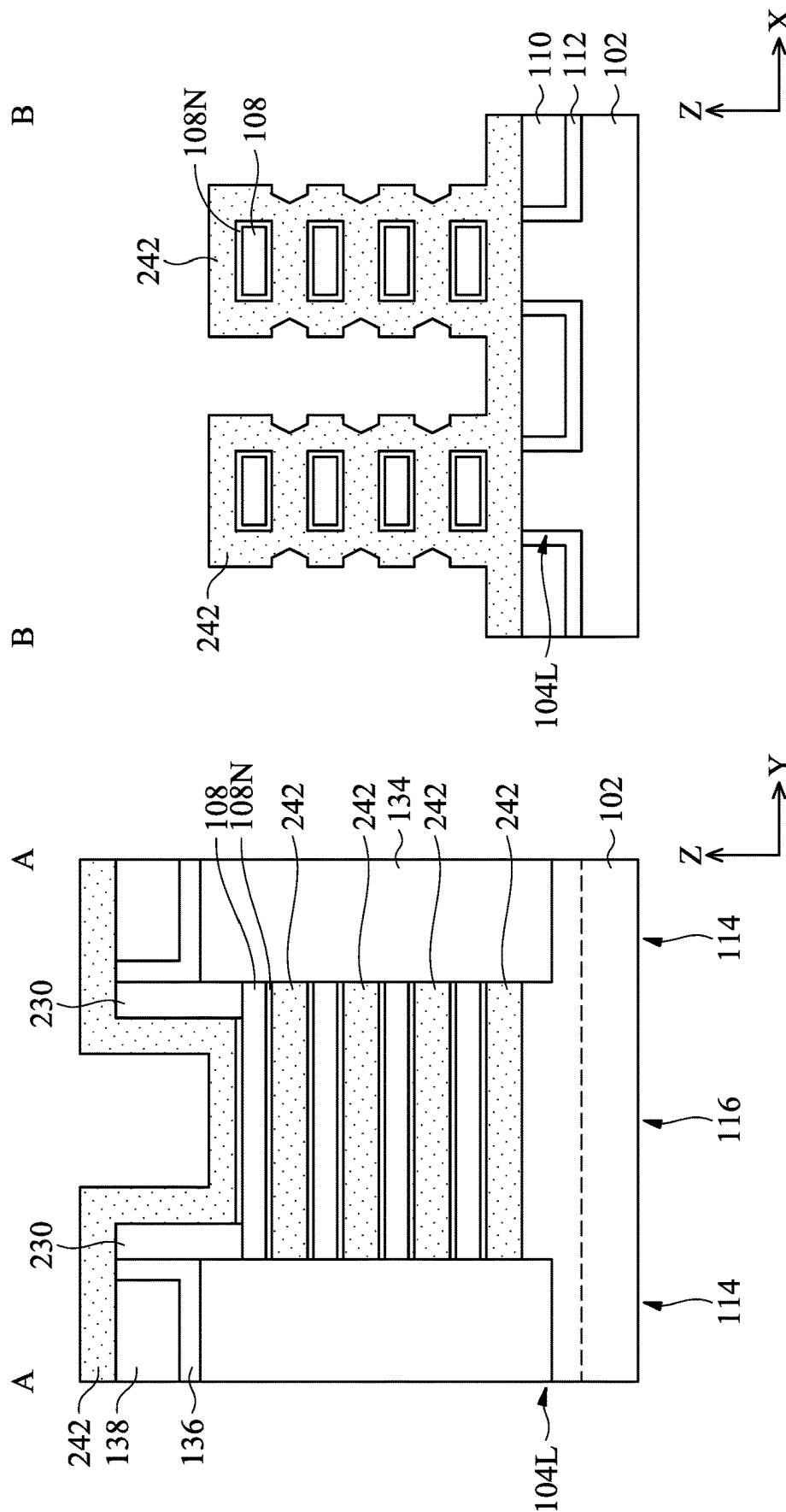


FIG. 2P-2

FIG. 2P-1



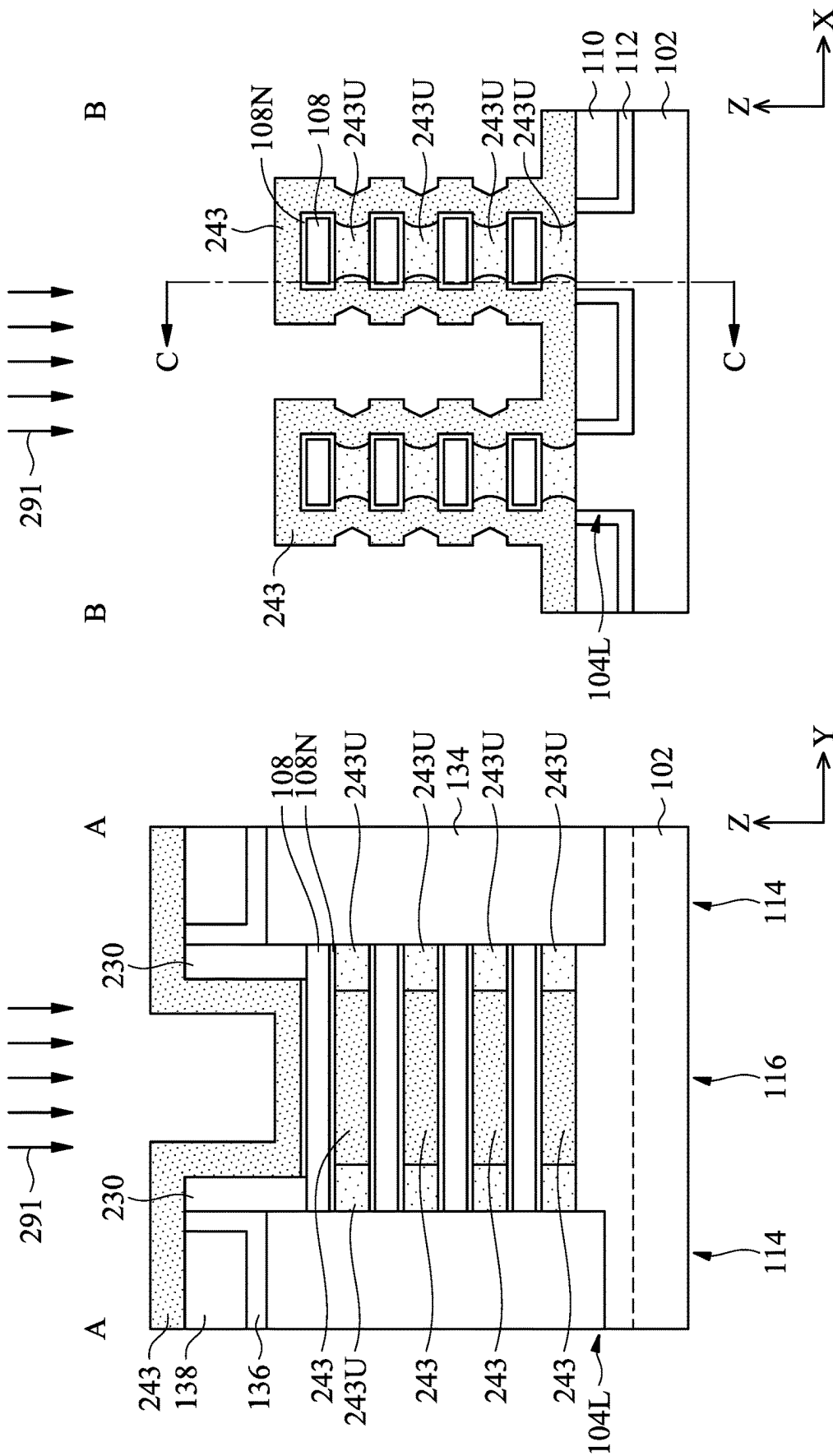


FIG. 3B-1

FIG. 3B-2

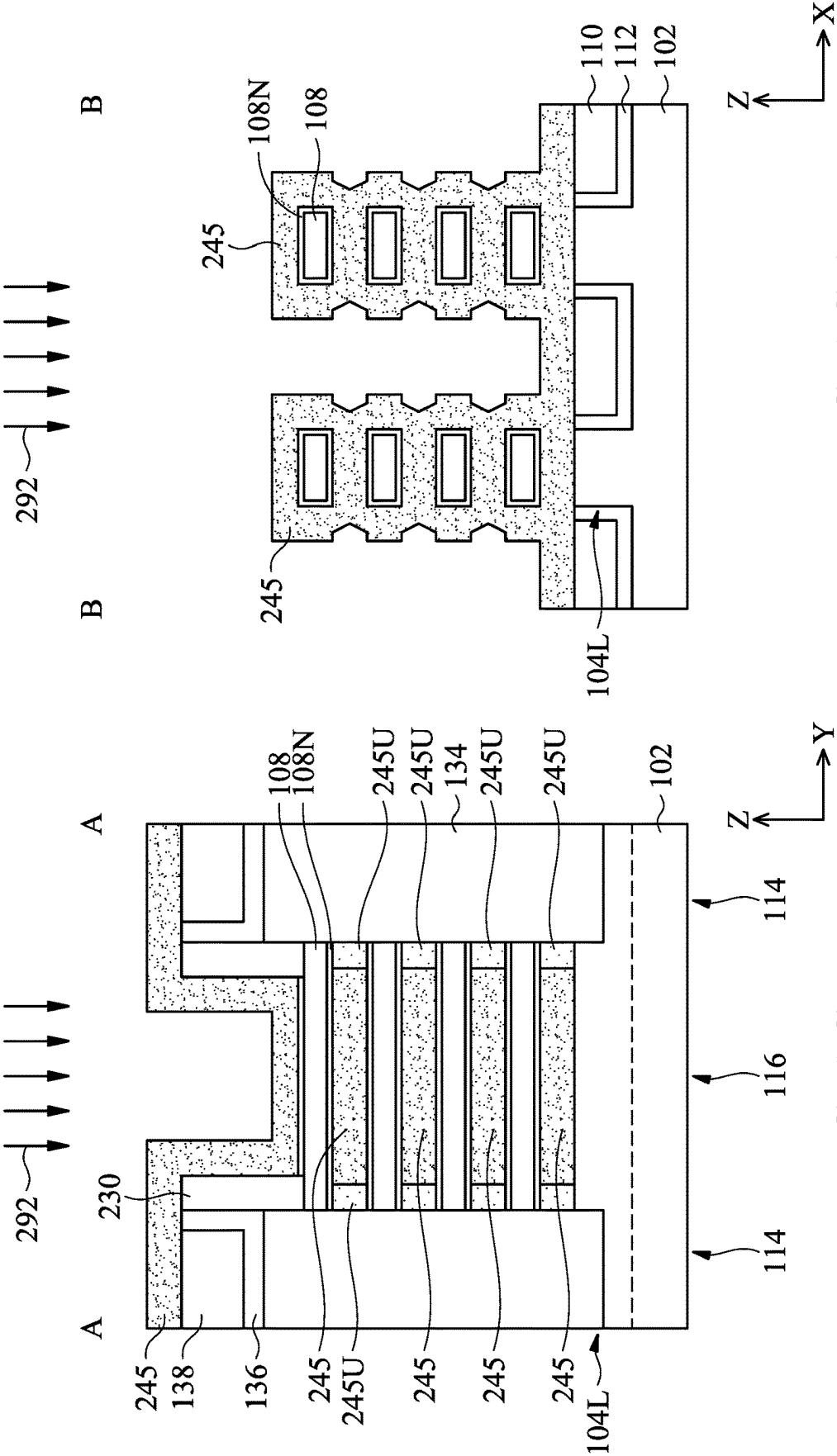


FIG. 3C-1

FIG. 3C-2

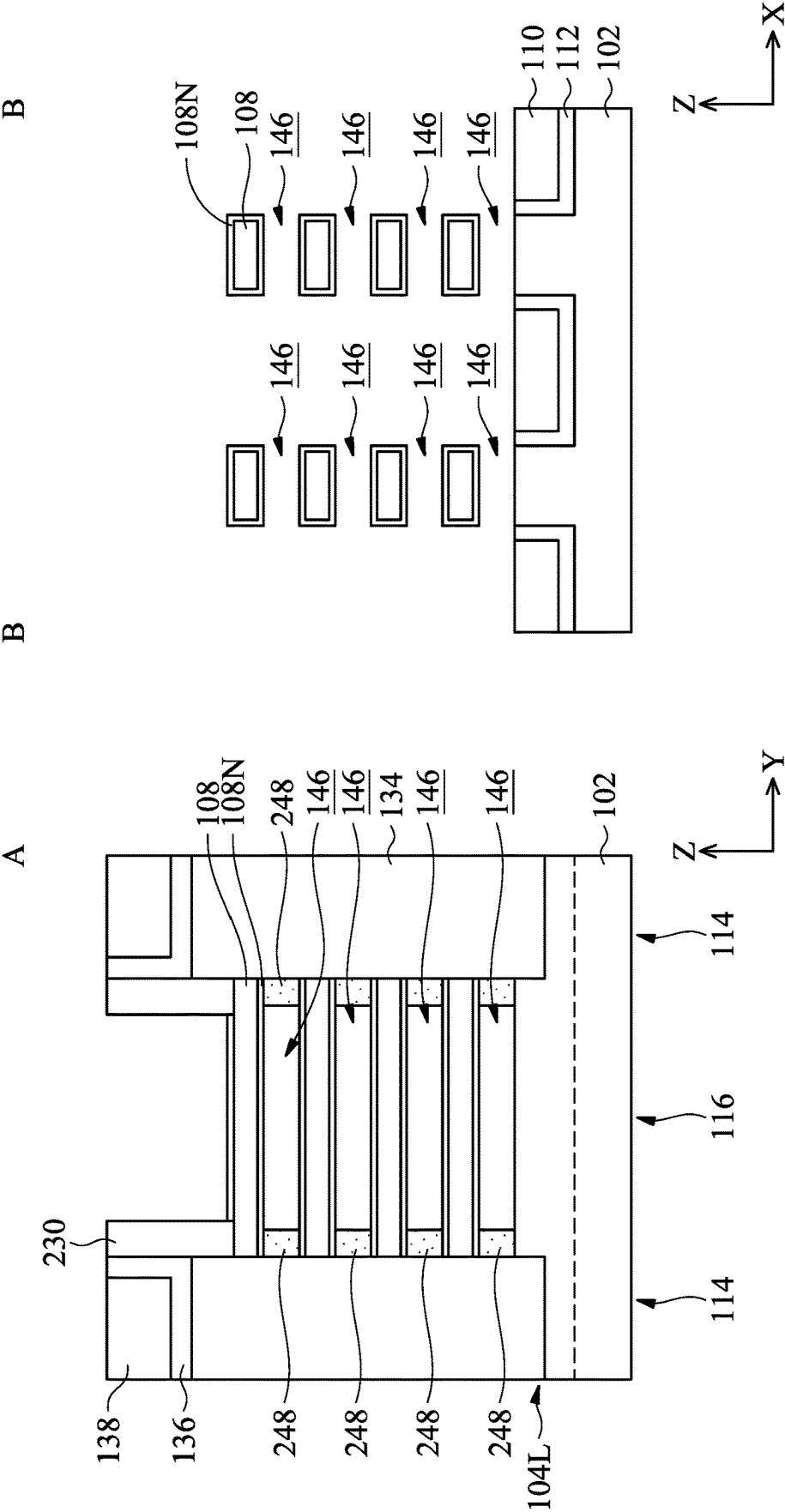


FIG. 3D-1

FIG. 3D-2

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SEMICONDUCTOR DEVICE STRUCTURE WITH INNER SPACER LAYER

PRIORITY CLAIM

This application is a Continuation Application of U.S. application Ser. No. 18/182,774, filed on Mar. 13, 2023, entitled “SEMICONDUCTOR DEVICE STRUCTURE WITH INNER SPACER LAYER AND METHOD FOR FORMING THE SAME,” which is a Continuation Application of U.S. application Ser. No. 17/504,104 (now U.S. Pat. No. 11,605,728), filed on Oct. 18, 2021, entitled “SEMICONDUCTOR DEVICE STRUCTURE WITH INNER SPACER LAYER,” which is a Divisional Application of U.S. application Ser. No. 16/299,531 (now U.S. Pat. No. 11,152,491), filed on Mar. 12, 2019, entitled “METHOD FOR FORMING SEMICONDUCTOR DEVICE STRUCTURE WITH INNER SPACER LAYER,” which claims the benefit of U.S. Provisional Application No. 62/721,931, filed on Aug. 23, 2018 and entitled “NANO WIRE/SHEET DEVICE STRUCTURE AND METHOD FOR FORMING THE SAME,” all of which are incorporated herein by reference.

BACKGROUND

The electronics industry has experienced an ever-increasing demand for smaller and faster electronic devices which are simultaneously able to support a greater number of increasingly complex and sophisticated functions. Accordingly, there is a continuing trend in the semiconductor industry to manufacture low-cost, high-performance, and low-power integrated circuits (ICs). So far, these goals have been achieved in large part by scaling down semiconductor IC dimensions (e.g., minimum feature size) and thereby improving production efficiency and lowering associated costs. However, such miniaturization has introduced increased complexity into the semiconductor manufacturing process. Thus, the realization of continued advances in semiconductor ICs and devices calls for similar advances in semiconductor manufacturing processes and technology.

Recently, multi-gate devices have been introduced in an effort to improve gate control by increasing gate-channel coupling, reduce OFF-state current, and reduce short-channel effects (SCEs). One such multi-gate device that has been introduced is the gate-all around transistor (GAA). The GAA device gets its name from the gate structure which can extend around the channel region providing access to the channel on two or four sides. GAA devices are compatible with conventional complementary metal-oxide-semiconductor (CMOS) processes and their structure allows them to be aggressively scaled-down while maintaining gate control and mitigating SCEs. In conventional processes, GAA devices provide a channel in a silicon nanowire. However, integration of fabrication of the GAA features around the nanowire can be challenging. For example, while the current methods have been satisfactory in many respects, continued improvements are still needed.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying Figures. It should be noted that, in accordance with the standard practice in the industry, various

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features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 is a perspective view of a semiconductor device structure, in accordance with some embodiments of the disclosure.

FIGS. 2A-1 through 2P-1 are cross-sectional views illustrating the formation of a semiconductor device along line A-A in FIG. 1 at various intermediate stages, in accordance with some embodiments.

FIGS. 2A-2 through 2P-2 are cross-sectional views illustrating the formation of a semiconductor device along line B-B in FIG. 1 at various intermediate stages, in accordance with some embodiments.

FIGS. 3A-1 through 3D-1 are cross-sectional views illustrating the formation of an inner spacer layer along line A-A in FIG. 1 at various intermediate stages, in accordance with some embodiments.

FIGS. 3A-2 through 3D-2 are cross-sectional views illustrating the formation of an inner spacer layer along line B-B in FIG. 1 at various intermediate stages, in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the subject matter provided. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Some variations of the embodiments are described. Throughout the various views and illustrative embodiments, like reference numerals are used to designate like elements. It should be understood that additional operations can be provided before, during, and after the method, and some of the operations described can be replaced or eliminated for other embodiments of the method.

The gate all around (GAA) transistor structures described below may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

Embodiments for forming a semiconductor device (e.g. GAA structure) are provided. The method for forming the semiconductor device may include forming an inner spacer

layer between a gate stack and a source/drain feature to reduce the parasitic capacitance between the gate stack and the source/drain feature (i.e. C_{gs} and C_{gd}). In addition, the formation of the inner spacer layer may include forming a dielectric material followed by locally treating the dielectric material. Because an etching selectivity exists between the treated portion and the untreated portion of the dielectric material, a subsequent etching process can be well controlled to remove the treated portion thereby leaving the untreated portion to serve as an inner spacer layer.

FIG. 1 is a perspective view of a semiconductor device structure **50**, in accordance with some embodiments of the disclosure. FIGS. 2A-1 through 2P-1 are cross-sectional views illustrating the formation of a semiconductor device **100** along line A-A in FIG. 1 at various intermediate stages, in accordance with some embodiments. FIGS. 2A-2 through 2P-2 are cross-sectional views illustrating the formation of the semiconductor device **100** along line B-B in FIG. 1 at various intermediate stages, in accordance with some embodiments.

A semiconductor device structure **50** is provided, as shown in FIG. 1, in accordance with some embodiments. The formation of the semiconductor device structure **50** includes providing a substrate **102**, and forming fin structures **104** and isolation structures **110** over the substrate **102**, in accordance with some embodiments. The fin structures **104** are separated from each other by the isolation structure **110**, in accordance with some embodiments. Each of the fin structures **104** is surrounded by the isolation structures **110**, in accordance with some embodiments.

In some embodiments, the substrate **102** is a semiconductor substrate such as a silicon substrate. In some embodiments, the substrate **102** includes an elementary semiconductor such as germanium; a compound semiconductor such as gallium nitride (GaN), silicon carbide (SiC), gallium arsenide (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), and/or indium antimonide (InSb); an alloy semiconductor such as SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or a combination thereof. In some embodiments, the substrate **102** includes an epitaxial layer (epi-layer) overlying a bulk semiconductor substrate. In some embodiments, the substrate **102** is a semiconductor-on-insulator (SOI) substrate which may include a semiconductor substrate, a buried oxide layer over the substrate, and a semiconductor layer over the buried oxide layer.

The fin structures **104** are arranged in the X direction, as shown in FIG. 1, in accordance with some embodiments. The fin structures **104** extend in the Y direction, in accordance with some embodiments. The fin structures **104** each include a lower portion **104L** and an upper portion **104U**, in accordance with some embodiments. The lower portion **104L** of the fin structure **104** is formed by a portion of the substrate **102**, in accordance with some embodiments. The upper portion **104U** of the fin structure **104** is formed by a stacked semiconductor structure, which includes first semiconductor layers **106** and second semiconductor layers **108** alternately stacked over the lower portion **104L**, in accordance with some embodiments. In some embodiments, there are between 2 and 10 first semiconductor layers **106** and there are between 2 and 10 second semiconductor layers **108**.

As explained in detail below, the first semiconductor layers **106** of the fin structures **104** will be removed so that the second semiconductor layers **108** of the fin structures **104** form a nanowire structure extending between source/drain features, in accordance with some embodiments. The

nanowire structure of the second semiconductor layers **108** will be surrounded by a gate stack to serve as a channel region of the semiconductor device **100**, in accordance with some embodiments. For example, the embodiments described herein illustrate processes and materials that may be used to form nanowire structures with a GAA design for n-type FinFETs or p-type FinFETs.

The first semiconductor layers **106** and the second semiconductor layers **108** are made of materials having different lattice constants, in accordance with some embodiments. In some embodiments, the first semiconductor layers **106** and the second semiconductor layers **108** include Si, Ge, SiGe, GaAs, InSb, GaP, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb, or InP.

In some embodiments, the first semiconductor layers **106** are made of SiGe, where the percentage of germanium (Ge) in the SiGe is in the range from about 20 atomic % to about 50 atomic %, and the second semiconductor layers **108** are made of silicon. In some embodiments, the first semiconductor layers **106** are $\text{Si}_{1-x}\text{Ge}_x$, where x is more than about 0.3, or Ge ($x=1.0$) and the second semiconductor layers **108** are Si or $\text{Si}_{1-y}\text{Ge}_y$, where y is less than about 0.4, and $x>y$.

In some embodiments, the thickness of each of the first semiconductor layers **106** is in a range from about 1.5 nanometers (nm) to about 20 nm. In some embodiments, the first semiconductor layers **106** are substantially uniform in thickness. In some embodiments, the thickness of each of the second semiconductor layers **108** is in a range from about 1.5 nm to about 20 nm. In some embodiments, the second semiconductor layers **108** are substantially uniform in thickness.

In some embodiments, the formation of the fin structures **104** includes forming a stacked semiconductor structure including a first semiconductor material for first semiconductor layers **106** and a second semiconductor material for second semiconductor layers **108** over the substrate **102**, and patterning the stacked semiconductor structure and the underlying substrate **102**.

In some embodiments, the first semiconductor materials and the second semiconductor materials are formed using low pressure chemical vapor deposition (LPCVD), epitaxial growth process, another suitable method, and/or a combination thereof. In some embodiments, the epitaxial growth process includes molecular beam epitaxy (MBE), metal organic chemical vapor deposition (MOCVD), or vapor phase epitaxy (VPE).

In some embodiments, the patterning process includes forming a hard mask layer over the stacked semiconductor structure, and etching the semiconductor structure and the underlying substrate **102** through the hard mask layer. In some embodiments, the hard mask layer includes one or more dielectric material, such as silicon oxide, silicon nitride, silicon oxynitride, other suitable dielectric materials, and/or a combination thereof.

In some embodiments, the etching process of the patterning process removes the stacked semiconductor structure which is uncovered by the hard mask layer and further recesses the substrate **102** to form trenches. In some embodiments, after the etching process, the substrate **102** has portions which protrudes from between the trenches to form the lower portions **104L** of the fin structure **104**. In some embodiments, a remaining stacked semiconductor structure forms the upper portion **104U** of the fin structure **104**. In some embodiments, the etching process includes a dry etching process, such as reactive ion etch (RIE) or neutral beam etch (NBE), a wet etching process, and/or a combination thereof.

After the fin structures **104** are formed, a lining layer **112** is conformally formed along the substrate **102**, the fin structures **104**, and the hard mask layer, in accordance with some embodiments. In some embodiment, the lining layer **112** includes a bilayer structure, such as a silicon oxide layer and a silicon nitride layer formed on the silicon oxide layer. In some embodiments, the lining layer **112** is formed using a thermal oxidation, CVD, atomic layer deposition (ALD), another suitable method, and/or a combination thereof.

Afterward, an insulating material for the isolation structures **110** is formed over the lining layer **112**, in accordance with some embodiments. The insulating material fills the trenches and covers the upper surface of the hard mask layer, in accordance with some embodiments.

In some embodiments, the insulating material includes silicon oxide, silicon nitride, silicon oxynitride (SiON), other suitable insulating materials, and/or a combination thereof. In some embodiments, the insulating material is formed using LPCVD, plasma enhanced CVD (PECVD), high density plasma CVD (HDP-CVD), high aspect ratio process (HARP), flowable CVD (FCVD), ALD, another suitable method, and/or a combination thereof.

Afterward, the insulating material and the lining layer **112** formed above the hard mask layer are removed, in accordance with some embodiments. In some embodiments, the removal process is an etch-back process, chemical mechanical polishing (CMP), and/or a combination thereof. In some embodiments, the removal process also removes the hard mask layer and exposes the upper surfaces of the fin structures **104**.

The isolation structures **110** and the lining layer **112** are recessed to form the trenches **111**, as shown in FIGS. 2A-1 and 2A-2, in accordance with some embodiments. In some embodiments, the isolation structures **110** and the lining layer **112** are recessed using one or more selective etch processes, such as a dry etching process, a wet etching process, and/or a combination thereof.

The trenches **111** expose the upper surfaces and the sidewalls of the upper portions **104U** of the fin structures **104**, in accordance with some embodiments. The upper portion **104U** of each of the fin structures **104** protrudes from between the recessed isolation structures **110**, in accordance with some embodiments. The lower portions **104L** of the fin structures **104** are embedded in the recessed isolation structures **110**, in accordance with some embodiments.

A dummy gate structure **118** and a hard mask layer **124** are formed over the fin structures **104**, as shown in FIGS. 2B-1 and 2B-2, in accordance with some embodiments. The hard mask layer **124** is formed over the dummy gate structure **118**, in accordance with some embodiments. The dummy gate structure **118** extends in the X direction and across the fin structures **104**, in accordance with some embodiments.

The dummy gate structure **118** is used to define source/drain regions **114** of the fin structure **104** and a channel region **116** of the fin structure **104**, as shown in FIG. 2B-1, in accordance with some embodiments. The source/drain regions **114** are on opposite sides of the channel region **116**, in accordance with some embodiments. In specific, the dummy gate structure **118** is filled into the recesses **111** (shown in FIG. 2A-2) and covers the upper surfaces and the sidewalls of the fin structures **104** in the channel region **116**, in accordance with some embodiments. The dummy gate structure **118** does not cover the fin structures in the source/drain regions **114**, in accordance with some embodiments.

The dummy gate structure **118** includes a dummy gate dielectric layer **120** and a dummy gate electrode layer **122**,

in accordance with some embodiments. In some embodiments, the dummy gate dielectric layer **120** is made of one or more dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride (SiON), and/or a combination thereof. In some embodiments, the dummy gate dielectric layer **120** is made of a dielectric material with high dielectric constant (k value), for example, greater than 3.9. In some embodiments, the high-K dielectric material includes HfO₂, HfZrO, HfSiO, HfTiO, HfAlO, another suitable high-K dielectric material, and/or a combination thereof. In some embodiments, the dielectric material is formed using a thermal oxidation, CVD, ALD, physical vapor deposition (PVD), another suitable method, and/or a combination thereof.

In some embodiments, the dummy gate electrode layer **122** is made of a conductive material. In some embodiments, the conductive material includes polycrystalline-silicon (poly-Si), poly-crystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metals, and/or a combination thereof. In some embodiments, the conductive material is formed using CVD, PVD, and/or a combination thereof.

In some embodiments, the formation of the dummy gate structure **118** includes conformally forming the dielectric material for the dummy gate dielectric layer **120** along the upper surface of the substrate **102** and the sidewalls and the upper surfaces of the upper portions **104U** of the fin structures **104**, and forming the conductive material for dummy gate electrode layer **122** over the dielectric material for the dummy gate dielectric layer **120**.

In some embodiments, the hard mask layer **124** is formed over the conductive material for the dummy gate electrode layer **122**. In some embodiments, the hard mask layer **124** is used as an etching mask for forming the dummy gate structure **118**. In some embodiments, the formation of the hard mask layer **124** includes forming an oxide layer **126** (e.g., silicon oxide) over the conductive material for dummy gate electrode layer **122**, forming a nitride layer **128** (e.g., silicon nitride) over the oxide layer **126**, and patterning the oxide layer **126** and the nitride layer **128** using photolithography and etching processes.

In some embodiments, the dielectric material for the dummy gate dielectric layer **120** and the conductive material for dummy gate electrode layer **122**, uncovered by hard mask layer **124**, are removed using one or more etching processes, thereby exposing the fin structures **104** in the source/drain regions **114**. In some embodiments, the one or more etching processes are dry etching processes, wet etching processes, or a combination thereof.

A gate spacer layer **130** is formed over the substrate **102**, as shown in FIGS. 2C-1 and 2C-2, in accordance with some embodiments. The gate spacer layer **130** is conformally formed along the upper surfaces of the fin structures **104** in the source/drain regions **114**, the sidewalls of the dummy gate structure **118**, and the sidewalls and the upper surface of the hard mask layer **124**, in accordance with some embodiments.

In some embodiments, the gate spacer layer **130** is made of a dielectric material, such as silicon oxide (SiO₂), silicon nitride (SiN), silicon carbide (SiC), silicon oxynitride (SiON), silicon carbon nitride (SiCN), silicon oxide carbonitride (SiOCN), and/or a combination thereof. In some embodiments, the gate spacer layer **130** is formed using CVD (such as LPCVD, PECVD, or sub-atmospheric CVD (SACVD)), ALD, another suitable method, and/or a combination thereof.

The gate spacer layer **130** formed along the upper surfaces of hard mask layer **124** and the fin structures **104** is removed, as shown in FIGS. **2D-1** and **2D-2**, in accordance with some embodiments. After the removal process, the gate spacer layer **130**, formed along the sidewalls of the dummy gate structure **118** and the hard mask layer **124**, remains unre-
 5 removed and forms a pair of gate spacer layers **130** on the opposite sides of the dummy gate structure **118**, in accordance with some embodiments. The removal process includes a dry etching process, a wet etching process, and/or
 10 a combination thereof.

The fin structures **104** in the source/drain regions **114** are recessed to form source/drain recesses **132**, as shown in FIGS. **2E-1** and **2E-2**, in accordance with some embodiments. The source/drain recesses **132** is formed by etching the fin structures **104** uncovered by the gate spacer layers **130**, the dummy gate structure **118**, and the hard mask layer **124**, in accordance with some embodiments. The source/
 15 drain recesses **132** pass through the upper portions **104U** of the fin structures **104** and extend into the lower portion **104L** of the fin structures **104**, in accordance with some embodiments. In some embodiments, the etching process includes a dry etching process, a wet etching process, and/or a
 20 combination thereof.

Source/drain features **134** are formed in the respective source/drain recesses **132**, as shown in FIGS. **2F-1** and **2F-2**, in accordance with some embodiments. The source/drain features **134** are formed over the exposed upper surface of the lower portions **104L** of the fin structures **104**, in accordance with some embodiments. The portion of the fin
 25 structure **104** covered by the gate spacer layers **130** and the dummy gate structure **118** is sandwiched between the source/drain features **134**, in accordance with some embodiments.

In some embodiments, the source/drain features **134** are made of Ge, Si, GaAs, AlGaAs, SiGe, GaAsP, SiP, SiC, SiCP, another suitable material, and/or a combination thereof. In some embodiments, the source/drain features **134** are formed using epitaxial growth process, such as MBE, MOCVD, VPE, another suitable epitaxial growth process, and/or a combination thereof.
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In some embodiments, the source/drain features **134** are in-situ doped during the epitaxial growth process. For example, the source/drain features **134** may be the epitaxially grown SiGe doped with boron (B). For example, the source/drain features **134** may be the epitaxially grown Si doped with carbon to form silicon:carbon (Si:C) source/
 40 drain features, phosphorous to form silicon:phosphor (Si:P) source/drain features, or both carbon and phosphorous to form silicon carbon phosphor (SiCP) source/drain features. In some embodiments, the source/drain features **134** are doped in one or more implantation processes after the epitaxial growth process.

In some embodiments, the source/drain features **134** are activated by an annealing process. In some embodiments, the annealing processes include a rapid thermal annealing (RTA), a laser annealing process, other suitable annealing process, and/or a combination thereof.

A contact etching stop layer (CESL) **136** is formed over the substrate **102**, as shown in FIGS. **2G-1** and **2G-2**, in accordance with some embodiments. The CESL **136** is conformally formed along the upper surface of the source/
 45 drain features **134**, the sidewalls and the upper surfaces of the gate spacer layers **130**, and the upper surface of the hard mask layer **124**, in accordance with some embodiments.

In some embodiments, the CESL **136** is made of a dielectric material, such as silicon nitride, silicon oxide,

silicon oxynitride, another suitable dielectric material, or a combination thereof. In some embodiments, the CESL **136** is formed using CVD (such as PECVD, HARP, and/or a combination thereof), ALD, another suitable method, and/or
 5 a combination thereof.

An interlayer dielectric (ILD) layer **138** is formed over the CESL **136**, as shown in FIGS. **2H-1** and **2H-2**, in accordance with some embodiments. In some embodiments, the ILD layer **138** is made of a dielectric material, such as tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass (USG), or doped silicon oxide such as borophosphosilicate glass (BPSG), fluoride-doped silicate glass (FSG), phospho-
 10 silicate glass (PSG), borosilicate glass (BSG), and/or another suitable dielectric material. In some embodiments, the dielectric material for the ILD layer **138** is formed over the CESL **136** using CVD (such as HDP-CVD, PECVD, or HARP), ALD, another suitable method, and/or a combina-
 15 tion thereof.

Afterward, the dielectric material for the ILD layer **138**, the CESL **136** and the hard mask layer **124** above the dummy gate structure **118** are planarized using such as CMP process or an etch-back process, in accordance with some
 20 embodiments. After the planarization process, the upper surface of the dummy gate electrode layer **122** is exposed, in accordance with some embodiments.

The dummy gate structure **118** including the dummy gate electrode layer **122** and the dummy gate dielectric layer **120** is removed, as shown in FIGS. **2I-1** and **2I-2**, in accordance with some embodiments. After the dummy gate structure **118** is removed, the upper surfaces and the sidewalls of the upper portions **104U** of the fin structures **104** in the channel regions **116** are exposed, in accordance with some embodi-
 25 ments.

In some embodiments, the removal process includes one or more etching processes. For example, when the dummy gate electrode layer **122** is polysilicon, a wet etchant such as a TMAH solution may be used to selectively remove the dummy gate electrode layer **122**. For example, the dummy gate dielectric layer **120** may be thereafter removed using a plasma dry etching, a dry chemical etching, and/or a wet etching.

The first semiconductor layers **106** of the fin structures **104** are removed to form gaps **140**, as shown in FIGS. **2J-1** and **2J-2**, in accordance with some embodiments. The gaps **140** are formed in the channel region **116** and extend between the neighboring second semiconductor layers **108** and between the lowermost second semiconductor layer **108** and the lower portion **104L** of the fin structure **104**, in
 30 accordance with some embodiments. The gaps **140** laterally extend directly below the gate spacer layers **130**, in accordance with some embodiments.

After the removal process, four main surfaces (an upper surface, two side surfaces, and a bottom surface) of each of the second semiconductor layers **108** are exposed, in accordance with some embodiments. The exposed second semiconductor layers **108** form a nanowire structure which will be surrounded by a gate stack, in accordance with some
 35 embodiments.

In some embodiments, the removal process includes a selective wet etching process, such as APM (e.g., ammonia hydroxide-hydrogen peroxide-water mixture) etching process. In some embodiments, the wet etching process uses etchants such as ammonium hydroxide (NH₄OH), tetramethylammonium hydroxide (TMAH), ethylenediamine pyrocatechol (EDP), and/or potassium hydroxide (KOH) solutions.

Native oxide layers **108N** are formed on the main surfaces of the second semiconductor layers **108**, as shown in FIGS. **2J-1** and **2J-2**, in accordance with some embodiments. For example, during the removal process of the first semiconductor layers **106**, the second semiconductor layers **108** may be oxidized to form the native oxide layers **108N**. For example, after the removal process of the first semiconductor layers **106**, the second semiconductor layers **108** may be oxidized to form the native oxide layers **108N** because the second semiconductor layers **108** are exposed under an environment containing O_2 or H_2O .

A dielectric material **142** is formed over the substrate **102**, as shown in FIGS. **2K-1** and **2K-2**, in accordance with some embodiments. The dielectric material **142** is conformally formed along the upper surface of the ILD layer **138**, the upper surface of the CESL **136**, the upper surfaces and the sidewalls of the gate spacer layers **130**, and the upper surface of the isolation structures **110**, in accordance with some embodiments. The dielectric material **142** is also formed over the native oxide layers **108N** and surrounds the four main surfaces of each of the second semiconductor layers **108**, in accordance with some embodiments. The gaps **140** (shown in FIGS. **2J-1** and **2J-2**) are entirely filled by the dielectric material **142**, in accordance with some embodiments.

The dielectric material **142** is made of a dielectric material with k-value lower than 6, such as in a range from about 2.0 to about 6.0, in accordance with some embodiments. The dielectric material **142** is used to form inner spacer layers to reduce the parasitic capacitance between the gate stack and the source/drain feature, in accordance with some embodiments. If the k-value of the dielectric material **142** is too high, the dielectric material may not help reduce the parasitic capacitance.

In some embodiments, the dielectric material **142** is silicon oxycarbide (SiOC), silicon oxide carbonitride (SiOCN), silicon carbon nitride (SiCN), and/or a combination thereof, in accordance with some embodiments. For example, the dielectric material **142** may be represented as $Si_wO_xC_yN_z$, where w, x, y and z < 1 and are the atomic percentages of Si, O, C and N, respectively. In some embodiments, the dielectric material **142** is formed using CVD (such as LPCVD, PECVD, SACVD, HARP, and/or FCVD), ALD, another suitable method, and/or a combination thereof.

The dielectric material **142** has a lower carbon concentration than the gate spacer layer **130**, in accordance with some embodiments. In some embodiments, the carbon concentration the dielectric material **142** is at least about 2% lower than the carbon concentration of the gate spacer layer **130**, for example, about 3%, or about 5% less. Due to the differentiated carbon concentration, the gate spacer layers **130** have little loss in a subsequent etching process for forming an inner spacer, in accordance with some embodiments.

For example, the gate spacer layer **130** may be silicon oxide carbonitride (SiOCN) and represented as the first material of $Si_wO_xC_yN_z$, where w is in a range from about 22.9 atomic % to about 32.9 atomic %; x is in a range from about 43.4 atomic % to about 53.4 atomic %; y is in a range from about 11.6 atomic % to about 21.6 atomic %; and z is in a range from about 3.1 atomic % to about 13.1 atomic %.

For example, the gate spacer layer **130** may be silicon oxide carbonitride (SiOCN) and represented as the second material of $Si_wO_xC_yN_z$, where w is in a range from about 27.3 atomic % to about 37.3 atomic %; x is in a range from about 26.7 atomic % to about 36.7 atomic %; y is in a range

from about 5.0 atomic % to about 17.0 atomic %; and z is in a range from about 19.0 atomic % to about 29.0%.

For example, the dielectric material **142** may be silicon oxide carbonitride (SiOCN) and represented as the third material of $Si_wO_xC_yN_z$, where w is in a range from about 45.9 atomic % to about 55.9 atomic %; x is in a range from about 22.5 atomic % to about 32.5 atomic %; y is in a range from about 0.1 atomic % to about 10.1 atomic %; and z is in a range from about 12.3 atomic % to about 22.3 atomic %.

For example, the dielectric material **142** may be silicon oxide carbonitride (SiOCN) and represented as the fourth material of $Si_wO_xC_yN_z$, where w is in a range from about 24.0 atomic % to about 34.0 atomic %; x is in a range from about 40.9 atomic % to about 50.9 atomic %; y is in a range from about 8.3 atomic % to about 18.3 atomic %; and z is in a range from about 6.8 atomic % to about 16.8 atomic %.

For example, the dielectric material **142** may be silicon oxycarbide (SiOC) and represented as the fifth material of $Si_wO_xC_yN_z$, where w is in a range from 24.0 atomic % to about 34.0 atomic %; x is in a range from about 55.2 atomic % to about 65.2 atomic %; y is in a range from about 5.1 atomic % to about 15.1 atomic %; and z is in a range from about 0.1 atomic % to about 5.0 atomic %.

For example, when the gate spacer layer **130** is the first material of SiOCN, the dielectric material **142** may be the third material, the fourth material, or the fifth material of SiOCN. For example, when the gate spacer layer **130** is the second material of SiOCN, the dielectric material **142** may be the third material or the fifth material of SiOCN.

In some embodiments, as measured in the X direction, the second semiconductor layer **108** has a dimension D1 in a range from about 5 nm to about 50 nm. In some embodiments, as measured in the Z direction, the second semiconductor layer **108** has a dimension D2 in a range from about 4 nm to about 10 nm. In some embodiments, as measured in the Z direction, the spacing between neighboring second semiconductor layers **108** is a dimension D3 in a range from about 1.5 nm to about 10 nm.

In some embodiments, as measured in the X direction, the dielectric material **142** formed along the side surface of the second semiconductor layer **108** has a maximum thickness D4 in a range from about 1 nm to about 5 nm. In some embodiments, as measured in the X direction, the dielectric material **142** formed between neighboring second semiconductor layers **108** has a minimum thickness D5 in a range from about 8 nm to about 16 nm.

In some embodiments, as measured in the Y direction, the dielectric material **142** formed along the sidewall of the gate spacer layer **130** has a thickness D6 in a range from about 1 nm to about 5 nm. In some embodiments, as measured in the Y direction, the spacing between the dielectric material **142** formed along the respective sidewalls of the opposite gate spacer layers **130** is a dimension D7 in a range from about 6 nm to about 20 nm.

FIGS. **2L-1**, **2L-2**, **2M-1**, **2M-2**, **2N-1** and **2N-2** illustrate treatment processes performed on the dielectric material **142**, in accordance with some embodiments. After the treatment processes, the dielectric material **142** has a treated portion **145** and an untreated portion **145U**, as shown in FIGS. **2N-1** and **2N-2**, in accordance with some embodiments. Because an etching selectivity exists between the treated portion **145** and the untreated portion **145U**, a subsequent etching process selectively removes the treated portion **145** of the dielectric material **142** and remains the untreated portion **143U** as inner spacer layers, in accordance with some embodiments. The treatment processes of the dielectric material **142** are described in detail below.

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The dielectric material **142** is treated using a first treatment process **191**, as shown in FIGS. **2L-1** and **2L-2**, in accordance with some embodiments. It should be noted that FIG. **2L-1** is taken along line C-C in FIG. **2L-2**. The outer portion of the dielectric material **142** is treated and is referred to as a first treated portion **143** while an inner portion of the dielectric material **142** remains untreated and is referred to as an untreated portion **143U**, in accordance with some embodiments.

The first treated portion **143**, which is formed in the channel region **116**, is in direct contact with the native oxide layers **108N**, as shown in FIG. **2L-2**, in accordance with some embodiments. The native oxide layer **108N**, which is formed on the upper surface and the side surfaces of the uppermost second semiconductor layer **108**, is entirely covered by the first treated portion **143**, in accordance with some embodiments. The native oxide layer **108N**, which is formed on the side surfaces of each of the lower three semiconductor layers **108**, is entirely covered by the first treated portion **143**, in accordance with some embodiments.

A peripheral area of the bottom surface of the uppermost second semiconductor layer **108** is covered by the first treated portion **143** while a center area of the bottom surface of the uppermost second semiconductor layer **108** is covered by the untreated portion **143U**, as shown in FIG. **2L-2**, in accordance with some embodiments. Peripheral areas of the upper surface and the bottom surface of each of the lower three semiconductor layers **108** are covered by the first treated portion **143** while center areas of the upper surface and the bottom surface of each of the lower three second semiconductor layers **108** are covered by the untreated portion **143U**, in accordance with some embodiments.

The first treated portion **143**, which is formed in the channel region **116**, extends between neighboring second semiconductor layers **108** and between the lowermost second semiconductor layer **108** and the lower portion **104L**, as shown in FIG. **2L-1**, in accordance with some embodiments. The dielectric material **142**, which is formed along the upper surfaces of the ILD layer **138** and the CESL **136**, and the upper surfaces and the sidewalls of the gate spacer layers **130**, is also treated and forms the first treated portion **143**, as shown in FIG. **2L-1**, in accordance with some embodiments.

Because of the shielding of the gate spacer layers **130**, the untreated portion **143U** remains in the source/drain regions **114** and is located on the source/drain features **134**, in accordance with some embodiments.

The first treatment process **191** oxidizes the dielectric material **142** to consume carbon in the dielectric material **142** and increase oxygen in the dielectric material **142**, in accordance with some embodiments. The first treatment process **191** is a remote plasma process, which is performed in an etching tool equipped with a remote plasma system (RPS), in accordance with some embodiments.

The first treatment process **191** uses oxidizing gases including H_2O and O_2 gases, in accordance with some embodiments. In some embodiments, a flow rate of H_2O is in a range from about 50 sccm to about 3000 sccm, and a flow rate of O_2 is in a range from about 50 sccm to about 3000 sccm. When passing through the RPS, the oxidizing gases may be formed into radicals. In some embodiments, the RF power of the plasma generator of the RPS is in a range from about 20 W to about 9000 W.

Afterward, the radicals are introduced into the etching chamber to the substrate **102** so as to oxidize the dielectric material **142**, in accordance with some embodiments. The charged species (such ion) generated from plasma generators are filtered using ground electrode material (such as

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Aluminum), ion trap material (such as Quartz), or another suitable material, and thus only radicals are formed in the etching chamber, in accordance with some embodiments. The first treatment process **191** treats the dielectric material **142** isotropically, in accordance with some embodiments. In some embodiments, the first treatment process **191** is performed in the etching chamber with a pressure ranging from about 0.1 Torr to about 15 Torr, and with a temperature in a range from about 20° C. to about 600° C. In some embodiments, the first treatment process **191** is performed for a first time period in a range from about 10 seconds to about 900 seconds.

During the first treatment process **191**, the first treated portion **143** is formed from the outer surface of the dielectric material **142** and grown into the interior of the dielectric material **142**, as shown in FIG. **2L-2**, in accordance with some embodiments. The first treatment process **191** is performed until the first treated portion **143** is grown to contact the native oxide layers **108N** in order to prevent further growth of the native oxide layers **108N** due to the first treatment process **191**, in accordance with some embodiments. Further growth of the native oxide layers **108N** would consume the second semiconductor layers **108**, resulting in shrinkage of the nanowire structure.

In some embodiments, as measured in the X direction, the first treated portion **143** has a thickness **D8** in a range from about 1 nm to about 5 nm. In some embodiments, thickness **D8** is substantially equal to thickness **D4**.

The dielectric material **142** and the native oxide layers **108N** are treated using a second treatment process **192**, as shown in FIGS. **2M-1** and **2M-2**, in accordance with some embodiments. It should be noted that FIG. **2M-1** is taken along line C-C in FIG. **2M-2**. The outer portion of the untreated portion **143U** is treated and forms a second treated portion **144** while the inner portion of the untreated portion **143U** remains untreated and is referred to as the untreated portion **144U**, as shown in FIG. **2M-2**, in accordance with some embodiments. The native oxide layers **108N** treated by the second treatment process **192** are labeled **108N'**.

A middle portion of the bottom surface of the uppermost second semiconductor layer **108** is covered by the second treated portion **144**, as shown in FIG. **2L-2**, in accordance with some embodiments. Middle portions the upper surface and the bottom surface of each of the lower three semiconductor layers **108** are covered by the second treated portion **144**, in accordance with some embodiments.

The second treated portion **144**, which is formed in the channel region **116**, extends between neighboring second semiconductor layers **108** and between the lowermost second semiconductor layer **108** and the lower portion **104L**, as shown in FIG. **2M-1**, in accordance with some embodiments.

Because of the shielding of the gate spacer layers **130**, the untreated portion **144U** remains in the source/drain regions **114** and is located on the source/drain features **134**, as shown FIG. **2M-1**, in accordance with some embodiments.

The second treatment process **192** nitridizes the dielectric material **142** and the native oxide layers **108N**, in accordance with some embodiments. The nitrided native oxide layers **108N'** will not be further grown in a following oxidation treatment process and serve as a passivation layer to prevent the second semiconductor layers **108** from being consumed in the following oxidation treatment process, in accordance with some embodiments. The second treatment process **192** is a remote plasma process, in accordance with some embodiments. For example, the second treatment

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process 192 may be performed in the same etching tool as the first treatment process 191.

The second treatment process 192 uses nitriding gas including NH_3 gas, in accordance with some embodiments. In some embodiments, a flow rate of NH_3 is in a range from about 50 sccm to about 3000 sccm. When passing through the RPS, the nitriding gas may be formed into radicals. In some embodiments, the RF power of the plasma generator of the RPS is in a range from about 20 W to about 9000 W.

Afterward, the radicals are introduced into the etching chamber to the substrate 102 so as to nitridize the dielectric material 142 and the native oxide layers 108N, in accordance with some embodiments. The charged species (such ion) generated from plasma generators are filtered using ground electrode material (such as Aluminum), ion trap material (such as Quartz), or another suitable material, and thus only radicals are formed in the etching chamber, in accordance with some embodiments. In some embodiments, the second treatment process 192 is performed in the etching chamber with a pressure ranging from about 0.1 Torr to about 15 Torr, and with a temperature in a range from about 20° C. to about 600° C. In some embodiments, the second treatment process 192 is performed for a second time period in a range from about 10 seconds to about 900 seconds.

In some embodiments, as measured in the X direction, the first treated portion 143 together with the second treated portion 144 has a thickness D9 in a range from about 2 nm to about 7 nm.

The dielectric material 142 is treated using a third treatment process 193, as shown in FIGS. 2N-1 and 2N-2, in accordance with some embodiments. The inner portion (including the second treated portion 144 and the untreated portion 144U (shown in FIG. 2M-2)) of the dielectric material 142 in the channel region 116 is entirely treated and the inner portion together with the first treated portion 143 are referred to as a treated portion 145, in accordance with some embodiments. Moreover, the first treated portion 143, formed along the upper surfaces of the ILD layer 138 and the CESL 136, and the upper surfaces and the sidewalls of the gate spacer layers 130, is also referred to as the treated portion 145.

The treated portion 145, which is formed in the channel region 116, extends between neighboring second semiconductor layers 108 and between the lowermost second semiconductor layer 108 and the lower portion 104L, as shown FIG. 2N-1, in accordance with some embodiments. In some embodiments, the treated portion 145 extends further into the source/drain regions 114 and directly below the gate spacer layer 130.

Because of the shielding of the gate spacer layers 130, the dielectric material 142, which is formed in the source/drain regions 114, remains untreated and is referred to as the untreated portion 145U, as shown FIG. 2N-1, in accordance with some embodiments. The untreated portion 145U is located on the source/drain feature 134, in accordance with some embodiments.

The third treatment process 193 oxidizes the dielectric material 142 to consume the carbon in the dielectric material 142 and increase oxygen in the dielectric material 142, in accordance with some embodiments. The third treatment process 193 is a remote plasma process, in accordance with some embodiments. For example, the third treatment process 193 may be performed in the same etching tool as the first treatment process 191.

The third treatment process 193 uses oxidizing gases including H_2O and O_2 gases, in accordance with some embodiments. The nitrided native oxide layers 108N' is not

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further grown in the third treatment process 193. In some embodiments, a flow rate of H_2O is in a range from about 50 sccm to about 3000 sccm, and a flow rate of O_2 is in a range from about 50 sccm to about 3000 sccm. When passing through the RPS, the oxidizing gases may be formed into radicals. In some embodiments, the RF power of the plasma generator of the RPS is in a range from about 20 W to about 9000 W.

Afterward, the radicals are introduced into the etching chamber to the substrate 102 so as to oxidize the dielectric material 142, in accordance with some embodiments. The charged species (such ion) generated from plasma generators are filtered using ground electrode material (such as Aluminum), ion trap material (such as Quartz), or another suitable material so that only radicals are formed in the etching chamber, in accordance with some embodiments. In some embodiments, the third treatment process 193 is performed in the etching chamber with a pressure ranging from about 0.1 Torr to about 15 Torr, and with a temperature in a range from about 20° C. to about 600° C. In some embodiments, the third treatment process 193 is performed for a third time period in a range from about 10 seconds to about 900 seconds. In order to treat the entire inner portion of the dielectric material 142 in the channel region 116, the RF power, temperature, pressure, and/or time period of the third treatment process 193 is greater than that of the first treatment process 191, in accordance with some embodiments.

The carbon in the dielectric material 142 is consumed by the oxidizing gases during the first treatment process 191 and third treatment process 193, thus the carbon concentration of the treated portion 145 is lower than that of the untreated portion 145U, in accordance with some embodiments. The oxygen concentration of the treated portion 145 is greater than that of the untreated portion 145U, in accordance with some embodiments. The treated portion 145 is a more oxide-like dielectric material compared to the untreated portion 145U, in accordance with some embodiments. In some embodiments, the carbon concentration of the treated portion 145 is at least about 3.0 atomic % lower than that of the untreated portion 145U, for example, about 5.0 atomic %, or about 10.0 atomic % less.

The treated portion 145 is removed to form gaps 146, as shown in FIGS. 2O-1 and 2O-2, in accordance with some embodiments. The removal process includes a selective etching process, in accordance with some embodiments. The selective etching process removes the treated portion 145 of the dielectric material 142 and remains the untreated portion 145U of the dielectric material 142 as inner spacer layers 148, in accordance with some embodiments.

The inner spacer layers 148 are formed in the source/drain regions 114 and on the source/drain features 134, in accordance with some embodiments. The inner spacer layers 148 are formed between neighboring second semiconductor layers 108 and between the lowermost second semiconductor layer 108 and the lower portion 104L, in accordance with some embodiments.

In some embodiments, as measured in the Y direction, the inner spacer layer 148 has a dimension D10 in a range from about 2.0 nm to about 9.0 nm. In some embodiments, as measured in the Y direction, the gate spacer layer 130 has a dimension D11 in a range from about 3.0 nm to about 16.0 nm. The gate spacer layer 130 laterally extends beyond an edge of the inner spacer layer 148 in the Y direction by a distance D12, in accordance with some embodiments. In some embodiments, the distance D12 is in a range from about 1.0 nm to about 7.0 nm.

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In some embodiments, the selective etching process of removing the treated portion **145** includes a wet etch process, a dry etch process, or a combination thereof. In some embodiments, the selective etching process is a plasma-free dry chemical etching process and isotropically etches the dielectric material **142**, in accordance with some embodiments. In some embodiments, the etchant of the dry chemical etching process includes radicals, such as HF, NF₃, NH₃, H₂, and/or a combination thereof.

Due to the differentiated carbon concentration, an etching selectivity exists between the treated portion **145** and the untreated portion **145U** in the etching process, in accordance with some embodiments. In some embodiments, the ratio of the etching rate of the treated portion **145** to the etching rate of the untreated portion **145U** is in a range from about 5 to about 100. As a result, the etching process can be well controlled to remove the treated portion **145** entirely and remain the untreated portion **145U** as the inner spacer layers **148**, in accordance with some embodiments.

In addition, in order to entirely remove the treated portion **145**, the etching process includes an over-etching step, in accordance with some embodiments. In the over-etching step, the etchants also etch the gate spacer layers **130**, in accordance with some embodiments. Due to the differentiated carbon concentration, an etching selectivity also exists between the treated portion **145** and the gate spacer layers **130** in the etching process, in accordance with some embodiments. In some embodiments, the ratio of the etching rate of the treated portion **145** to the etching rate of the gate spacer layer **130** is in a range from 5 to about 100. As a result, loss of the gate spacer layer **130** in the etching process of removing the treated portion **145** can be decreased, thereby maintaining the performance of the semiconductor device **100**, in accordance with some embodiments.

Afterward, the native oxide layers **108** and **108N'**, uncovered by the inner spacer layers **148**, are removed using an etching process, thereby exposing the main surfaces of the second semiconductor layers **108**, in accordance with some embodiments.

A gate stack **150** is formed in the channel region **116**, as shown in FIGS. 2P-1 and 2P-2, in accordance with some embodiments. The gate stack **150** extends in the X direction, in accordance with some embodiments. The gate stack **150** fills the gaps **146** (shown in FIGS. 2O-1 and 2O-2) and surrounds the second semiconductor layers **108**, in accordance with some embodiments. The gate stack **150** extends into the source/drain regions **114** and in contact with the inner spacer layers **148**, in accordance with some embodiments.

The gate stack **150** includes an interfacial layer **152**, a high-k dielectric layer **154**, and a gate electrode layer **156**, in accordance with some embodiments. The interfacial layer **152** is conformally formed along the main surfaces of the second semiconductor layers **108** to surround the second semiconductor layers **108**, in accordance with some embodiments. The interfacial layer **152** is conformally formed along the upper surface of the ILD layer **138**, the upper surface of the CESL **136**, the upper surfaces and the sidewalls of the gate spacer layers **130**, and the sidewalls of the inner spacer layers **148**, in accordance with some embodiments. In some embodiments, the interfacial layer **152** is made of a chemically formed silicon oxide.

The high-k dielectric layer **154** is formed on the interfacial layer **152**, in accordance with some embodiments. In some embodiments, the high-k gate dielectric layer **154** is made of one or more layers of a dielectric material, such as HfO₂, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, zirconium

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oxide, aluminum oxide, titanium oxide, hafnium dioxide-alumina (HfO₂—Al₂O₃) alloy, another suitable high-k dielectric material, and/or a combination thereof. In some embodiments, the high-k gate dielectric layer **154** is formed using CVD, ALD, another suitable method, and/or a combination thereof.

The gate electrode layer **156** is formed on the high-k dielectric layer **154**, in accordance with some embodiments. The gate electrode layer **156** fills the gaps **146**, in accordance with some embodiments. In some embodiments, the gate electrode layer **156** is made of one or more layers of conductive material, such as polysilicon, aluminum, copper, titanium, tantalum, tungsten, cobalt, molybdenum, tantalum nitride, nickel silicide, cobalt silicide, TiN, WN, TiAl, TiAlN, TaCN, TaC, TaSiN, metal alloys, another suitable material, and/or a combination thereof. In some embodiments, the gate electrode layer **156** is formed using CVD, ALD, electroplating, another suitable method, and/or a combination thereof.

Afterward, the gate structure **150**, which is formed above the upper surface of the ILD layer **138**, is planarized using such as CMP to expose the upper surface of the ILD layer **138**, in accordance with some embodiments. After the gate stack **150** is formed, the semiconductor device **100** is obtained.

The inner spacer layer **148**, formed between the source/drain feature **134** and the gate stack **150**, can reduce the parasitic capacitance between the gate stack and the source/drain feature (i.e. C_{gs} and C_{gd}), in accordance with some embodiments. As a result, the performance of the semiconductor device structure **100** can be increased.

In addition, the formation of the inner spacer layer **148** includes locally treating the dielectric material **142**, in accordance with some embodiments. Because an etching selectivity exists between the treated portion **145** and the untreated portion **145U** of the dielectric material **142**, the etching process can be well controlled to entirely remove the treated portion **145**, thereby leaving the untreated portion **145U** to serve as the inner spacer layers **148**. Without locally treating the dielectric material **142**, it will be difficult to control the formation of the inner spacer layers **148** on the desired region while entirely removing the dielectric material **142** formed on the semiconductor layers **108** and gate spacer layers **130**. Thus, the process window of the formation of the inner spacer layer **148** can be enhanced.

In the embodiments shown in FIGS. 2K-1 through 2O-2, the treatment processes **191**, **192** and **193** locally treat the dielectric material **142** into an oxide-like dielectric material. In the embodiments shown in FIGS. 3A-1 through 3D-2, treatment processes **291** and **292** locally treat a dielectric material **242** into a nitride-like dielectric material.

FIGS. 3A-1 through 3D-1 are cross-sectional views illustrating the formation of an inner spacer layer **248** along line A-A in FIG. 1 at various intermediate stages, in accordance with some embodiments. FIGS. 3A-2 through 3D-2 are cross-sectional views illustrating the formation of the inner spacer layer **248** along line B-B in FIG. 1 at various intermediate stages, in accordance with some embodiments.

Continuing from FIGS. 2J-1 and 2J-2, a dielectric material **242** is formed over the substrate **102**, as shown in FIGS. 3A-1 and 3A-2, in accordance with some embodiments. The dielectric material **242** is conformally formed along the upper surface of the ILD layer **138**, the upper surface of the CESL **136**, the upper surfaces and the sidewalls of the gate spacer layers **230**, and the upper surface of the isolation structures **110**, in accordance with some embodiments. The dielectric material **242** is formed over the native oxide layers

108N and surrounds the four main surfaces of each of the second semiconductor layers 108, in accordance with some embodiments. The gaps 140 (shown in FIGS. 2J-1 and 2J-2) are entirely filled by the dielectric material 242, in accordance with some embodiments.

The dielectric material 242 has a nitrogen concentration greater than the nitrogen concentration of the gate spacer layer 230, in accordance with some embodiments. In some embodiments, the nitrogen concentration of the dielectric material 242 is at least about 2% greater than the gate spacer layer 230, for example, about 3%, about 5% greater. Due to the differentiated nitrogen concentration, the gate spacer layer 230 has little loss in a subsequent etching process for forming an inner spacer, in accordance with some embodiments.

For example, the gate spacer layer 230 may be silicon carbon nitride (SiCN) and represented as the sixth material of $\text{Si}_w\text{O}_x\text{C}_y\text{N}_z$, where w is in a range from about 32.0 atomic % to about 42.0 atomic %; x is in a range from about 7.6 atomic % to about 17.6 atomic %; y is in a range from about 1.4 atomic % to about 11.4 atomic %; and z is in a range from about 39.0 atomic % to about 49.0 atomic %.

For example, the dielectric material 242 may be silicon carbon nitride (SiCN) and represented as the seventh material of $\text{Si}_w\text{O}_x\text{C}_y\text{N}_z$, where w is in a range from about 34.0 atomic % to about 44.0 atomic %; x is in a range from about 3.8 atomic % to about 13.8 atomic %; y is in a range from about 0.5 atomic % to about 6.0 atomic %; and z is in a range from about 45.5 atomic % to about 55.5 atomic %.

For example, when the gate spacer layer 230 is the sixth material of SiOCN, the dielectric material 242 may be the seventh material of SiOCN.

The dielectric material 242 is treated using a first treatment process 291, as shown in FIGS. 3B-1 and 3B-2, in accordance with some embodiments. It should be noted that FIG. 3B-1 is taken along line C-C in FIG. 3B-2. The outer portion of the dielectric material 242 is treated and is referred to as the first treated portion 243 while the inner portion of the dielectric material 242 remains untreated and is referred to as the untreated portion 243U, in accordance with some embodiments.

The profile of the first treated portion 243 is substantially the same as that of the first treated portion 143 as previously described and as illustrated in FIGS. 2L-1 and 2L-2, in accordance with some embodiments. For example, the first treated portion 243, which is formed in the channel region 116, is in direct contact with the native oxide layers 108N, as shown in FIG. 3B-2, in accordance with some embodiments. For example, because of the shielding of the gate spacer layers 230, the untreated portion 243U remains in the source/drain regions 114 and is located on the source/drain features 134, in accordance with some embodiments.

The first treatment process 291 nitridizes the dielectric material 242 to consume carbon in the dielectric material 242 and increase nitrogen in the dielectric material 242, in accordance with some embodiments. The first treatment process 291 is a remote plasma process, which is performed in an etching tool equipped with a remote plasma system (RPS), in accordance with some embodiments.

The first treatment process 291 uses nitriding gases including NH_3 gas, in accordance with some embodiments. In some embodiments, a flow rate of NH_3 is in a range from about 50 sccm to about 3000 sccm. When passing through the RPS, the nitriding gas may be formed into radicals. In some embodiments, a first RF power of the plasma generator of the RPS is in a range from about 20 W to about 9000 W.

Afterward, the radicals are introduced into the etching chamber to the substrate 102 so as to nitridize the dielectric material 242, in accordance with some embodiments. The charged species (such ion) generated from plasma generators are filtered using ground electrode material (such as Aluminum), ion trap material (such as Quartz), or another suitable material, and thus only radicals are formed in the etching chamber, in accordance with some embodiments. In some embodiments, the first treatment process 291 is performed in the etching chamber with a pressure ranging from about 0.1 Torr to about 15 Torr, and with a temperature in a range from about 20° C. to about 600° C. In some embodiments, the first treatment process 291 is performed for a first time period in a range from about 10 seconds to about 900 seconds.

The first treatment process 291 is performed until the first treated portion 243 is grown to contact the native oxide layers 108N in order to prevent the second semiconductor layers 108 from being nitrided, as shown in FIG. 3B-2, in accordance with some embodiments. For example, the nitridation of the second semiconductor layers 108 may consume the second semiconductor layers 108, resulting in shrinkage of the nanowire structure.

The dielectric material 242 is treated using a second treatment process 292, as shown in FIGS. 3C-1 and 3C-2, in accordance with some embodiments. The inner portion (including the untreated portion 243U (shown in FIG. 3B-2)) of the dielectric material 242 in the channel region 116 is entirely treated and the inner portion together with the first treated portion 243 are referred to as a treated portion 245, in accordance with some embodiments. Moreover, the first treated portion 243, formed along the upper surfaces of the ILD layer 138 and the CESL 136, and the upper surfaces and the sidewalls of the gate spacer layers 230, is also referred to as a treated portion 245.

The treated portion 245, which is formed in the channel region 116, extends between neighboring second semiconductor layers 108 and between the lowermost second semiconductor layer 108 and the lower portion 104L, as shown FIG. 3C-1, in accordance with some embodiments. In some embodiments, the treated portion 245 extends further into the source/drain regions 114 and directly below the gate spacer layer 230.

Because of the shielding of the gate spacer layers 230, the dielectric material 242, which is formed in the source/drain regions 114, remains untreated and is referred to as the untreated portion 245U. The untreated portion 245U is located on the source/drain feature 134, in accordance with some embodiments.

The second treatment process 292 nitridizes the dielectric material 242 to consume the carbon in the dielectric material 242 and increase nitrogen in the dielectric material 242, in accordance with some embodiments. The second treatment process 292 is a remote plasma process, in accordance with some embodiments. For example, the remote plasma process may be performed in the same etching tool as the first treatment process 291.

The second treatment process 292 uses nitriding gas including NH_3 gas, in accordance with some embodiments. In some embodiments, a flow rate of NH_3 is in a range from about 50 sccm to about 3000 sccm. When passing through the RPS, the nitriding gas may be formed into radicals. In some embodiments, the second RF power of the plasma generator of the RPS is in a range from about 20 W to about 9000 W. In order to treat the entire inner portion of the dielectric material 242 in the channel region 116 without nitriding the semiconductor layers 108, the RF power,

temperature, and/or pressure of the plasma generator of the RPS is lower than that of the first treatment process 291, in accordance with some embodiments.

Afterward, the radicals are introduced into the etching chamber to the substrate 102 so as to nitridize the dielectric material 242, in accordance with some embodiments. The charged species (such ion) generated from plasma generators are filtered using ground electrode material (such as Aluminum), ion trap material (such as Quartz), or another suitable material, and thus only radicals are formed in the etching chamber, in accordance with some embodiments. In some embodiments, the second treatment process 292 is performed in the etching chamber with a pressure ranging from about 0.1 Torr to about 15 Torr, and with a temperature in a range from about 20° C. to about 600° C. In some embodiments, the second treatment process 292 is performed for a second time period in a range from about 10 seconds to about 900 seconds. In order to treat the entire inner portion of the dielectric material 242 in the channel region 116, the second time period of the second treatment process 292 is greater than the first time period of the first treatment process 291, in accordance with some embodiments.

The nitrogen in the dielectric material 242 is increased by the nitriding gases during the first treatment process 291 and second treatment process 292, thus the nitrogen concentration of the treated portion 245 is greater than that of the untreated portion 245U, in accordance with some embodiments. The treated portion 245 is a more nitride-like dielectric material compared to the untreated portion 245U, in accordance with some embodiments. In some embodiments, the nitrogen concentration of the treated portion 245 is at least about 3.0 atomic % greater than that of the untreated portion 245U, for example, about 5.0 atomic %, or about 10.0 atomic % greater.

The treated portion 245 is removed to form gaps 146, as shown in FIGS. 3D-1 and 3D-2, in accordance with some embodiments. The removal process includes a selective etching process, in accordance with some embodiments. The selective etching process removes the treated portion 245 of the dielectric material 242 and remains the untreated portion 245U of the dielectric material 242 as inner spacer layers 248, in accordance with some embodiments.

In some embodiments, the selective etching process of removing the treated portion 245 includes a wet etch process, a dry etch process, or a combination thereof. In some embodiments, the selective etching process is a plasma-free dry chemical etching process and isotropically etches the dielectric material 242, in accordance with some embodiments. In some embodiments, the etchants of the dry chemical etching process includes radicals, such as HF, NH₃, NF₃, O₂, N₂O, H₂, Cl₂, Br₂, HCl, HBr, and/or a combination thereof.

Due to the differentiated nitrogen concentration, an etching selectivity exists between the treated portion 245 and the untreated portion 245U in the etching process, in accordance with some embodiments. In some embodiments, the ratio of the etching rate of the treated portion 245 to the etching rate of the untreated portion 245U is in a range from about 5 to about 100. As a result, the etching process can be well controlled to remove the treated portion 245 entirely and remain the untreated portion 245U as the inner spacer layer 248, in accordance with some embodiments.

In some embodiments, the ratio of the etching rate of the treated portion 245 to the etching rate of the gate spacer layer 230 is in a range from 5 to about 100. As a result, loss of the gate spacer layer 230 in the etching process of

removing the treated portion 245 can be decreased, thereby maintaining the performance of the semiconductor device, in accordance with some embodiments.

After the inner spacer layers 248 are formed, the processes that were described above and illustrated in FIGS. 2P-1 and 2P-2 may be performed to obtain the semiconductor device 100.

As described above, the inner spacer layer 148 (or 248), formed between the source/drain feature 134 and the gate stack 150, can reduce the parasitic capacitance between the gate stack and the source/drain feature. In addition, the formation of the inner spacer layer 148 (or 248) includes forming a dielectric material 142 (or 242) surrounding second semiconductor layers 108, and locally treating the dielectric material 142 (or 242) in the channel region 116. Because an etching selectivity exists between the treated portion 145 (or 245) and the untreated portion 145U (or 245U) of the dielectric material 142 (or 242), the etching process can be well controlled to entirely remove the treated portion 145 (or 245), thereby leaving the untreated portion 145U (or 245U) as the inner spacer layer 148 (or 248). As a result, the process window of formation of the inner spacer layer 148 (or 248) can be enhanced.

Embodiments of a method for forming a semiconductor device structure are provided. The method for forming the semiconductor device structure may include forming a dielectric material surrounding second semiconductor layers, and locally treating the dielectric material. After the dielectric material is treated, the treated portion of the dielectric material may be etched and the remaining portion of the dielectric material may form an inner spacer layer. Because the etching process is well controlled by locally treating the dielectric material, the process window of the formation of the inner spacer layer may be enhanced.

In some embodiments, a semiconductor device structure is provided. The semiconductor device structure includes a gate stack wrapping around a plurality of nanowire structures. The gate stack includes a first portion above the plurality of nanowire structures and second portions between the nanowire structures. The semiconductor device structure further includes a gate spacer layer along a sidewall of the first portion of the gate stack, and a plurality of inner spacer layers along sidewalls of the second portions of the gate stack. The gate spacer layer has a first carbon concentration, the inner spacer layers have a second carbon concentration, and the second carbon concentration is lower than the first carbon concentration.

In some embodiments, a method for forming a semiconductor device structure is provided. The method includes a nanowire structure and a source/drain feature over a fin structure, a gate stack surrounding the nanowire structures, a first spacer layer between the gate stack and the source/drain feature over the nanowire structures, and a second spacer layer between the gate stack and the source/drain feature under the nanowire structure. The first spacer layer has a first nitrogen concentration, the second spacer layer has a second nitrogen concentration, and the second nitrogen concentration is greater than the first nitrogen concentration.

In some embodiments, a semiconductor device structure is provided. The semiconductor device structure includes a plurality of nanowire structures, a gate stack wrapping around center portions of the plurality of nanowire structures, a plurality of oxide layers wrapping end portions of the plurality of nanowire structures, a source/drain feature adjoining the end portions of the plurality of nanowire

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structures, and a plurality of inner spacer layers between the plurality of oxide layers and between the source/drain feature and the gate stack.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor device structure, comprising:
 - a gate stack wrapping around a plurality of nanowire structures, wherein the gate stack includes a first portion above the plurality of nanowire structures and second portions between the nanowire structures;
 - a gate spacer layer along a sidewall of the first portion of the gate stack; and
 - a plurality of inner spacer layers along sidewalls of the second portions of the gate stack, wherein the gate spacer layer has a first carbon concentration, the inner spacer layers have a second carbon concentration, and the second carbon concentration is lower than the first carbon concentration.
2. The semiconductor device structure as claimed in claim 1, wherein the first portion of the gate stack is narrower than the second portions of the gate stack.
3. The semiconductor device structure as claimed in claim 1, wherein the inner spacer layers are narrower than the gate spacer layer.
4. The semiconductor device structure as claimed in claim 1, further comprising:
 - a source/drain feature adjoining the plurality of nanowire structures and the plurality of inner spacer layers.
5. The semiconductor device structure as claimed in claim 1, further comprising:
 - a plurality of oxide layers respectively between the plurality of inner spacer layers and the plurality of nanowire structures.
6. The semiconductor device structure as claimed in claim 5, wherein the oxide layers are in direct contact with the gate stack.
7. The semiconductor device structure as claimed in claim 1, wherein the first carbon concentration of the gate spacer layer is in a range from about 11.6 atomic % to about 21.6 atomic %.
8. The semiconductor device structure as claimed in claim 1, wherein the second carbon concentration of the inner spacer layers is in a range from about 5.1 atomic % to about 15.1 atomic %.
9. A semiconductor device structure, comprising:
 - a nanowire structure and a source/drain feature over a fin structure;
 - a gate stack surrounding the nanowire structures;
 - a first spacer layer between the gate stack and the source/drain feature over the nanowire structures; and
 - a second spacer layer between the gate stack and the source/drain feature under the nanowire structure, wherein the first spacer layer has a first nitrogen

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concentration, the second spacer layer has a second nitrogen concentration, and the second nitrogen concentration is greater than the first nitrogen concentration.

10. The semiconductor device structure as claimed in claim 9, wherein the first spacer layer has a first sidewall interfaced with the gate stack, the second spacer layer has a first sidewall interfaced with the gate stack, and the first sidewall of the second spacer layer is offset from the first sidewall of the first spacer.

11. The semiconductor device structure as claimed in claim 10, wherein the first spacer layer has a second sidewall interfaced with the source/drain feature, the second spacer layer has a second sidewall interfaced with the source/drain feature, and the second sidewall of the second spacer layer is substantially aligned with the second sidewall of the first spacer layer.

12. The semiconductor device structure as claimed in claim 9, wherein the first nitrogen concentration of the first spacer layer is in a range from about 39.0 atomic % to about 49.0 atomic %.

13. The semiconductor device structure as claimed in claim 9, wherein the second nitrogen concentration of the second spacer layers is in a range from about 45.5 atomic % to about 55.5 atomic %.

14. The semiconductor device structure as claimed in claim 9, wherein the source/drain feature includes a portion embedded in the fin structure.

15. The semiconductor device structure as claimed in claim 9, further comprising:

an isolation structure surrounding the fin structure.

16. A semiconductor device structure, comprising:

- a plurality of nanowire structures;
- a gate stack wrapping around center portions of the plurality of nanowire structures;
- a plurality of oxide layers wrapping end portions of the plurality of nanowire structures;
- a source/drain feature adjoining the end portions of the plurality of nanowire structures; and
- a plurality of inner spacer layers between the plurality of oxide layers and between the source/drain feature and the gate stack.

17. The semiconductor device structure as claimed in claim 16, wherein the plurality of oxide layers is in direct contact with the gate stack, the source/drain feature and the plurality of inner spacer layers.

18. The semiconductor device structure as claimed in claim 16, further comprising:

- a gate spacer layer above the plurality of nanowire structures and between the source/drain feature and the gate stack, wherein the gate spacer layer extends beyond sidewalls of the plurality of oxide layers interfaced with the gate spacer layer.

19. The semiconductor device structure as claimed in claim 18, wherein the gate spacer layer has a first carbon concentration, the plurality of inner spacer layers has a second carbon concentration, and the second carbon concentration is lower than the first carbon concentration.

20. The semiconductor device structure as claimed in claim 18, wherein the gate spacer layer has a first nitrogen concentration, the plurality of inner spacer layers has a second nitrogen concentration, and the second nitrogen concentration is greater than the first nitrogen concentration.